SolariNet User's Manual

How to use every screen and function of the SolariNet dashboard: monitoring and alerts, the status panel, inventory and labels, RackWire, discovery, provisioning, images and stacks, maintenance, Git, certificates and DNS.

FOR
The operator and any household member with dashboard access

SYSTEM
dashboard.akoria.net
SolariNet release on main, November 2026

EDITION
1.0 · 3 November 2026
US Letter · screen and print

Companion manuals. The *Developer's Manual* covers building, the C daemons and the code layout. The *API Expansion Manual* covers the HTTP API and how to extend it. This manual is about using the system, and it never contains secrets: credentials are named by file path only.

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CHAPTER 1

About SolariNet and this manual

SolariNet watches the Akoria home network and keeps its official record of systems, addresses and things owned. You use it through one web dashboard, a wall-mounted LED status panel and the messages it sends you.

1.1 What SolariNet does

SolariNet has five jobs. Each has its own part of the dashboard.

WATCH
Measures every enrolled host and probes every service from several vantage points. See Chapter 3.

WARN
Turns sustained problems into alerts, messages and a panel alarm, and stays quiet during maintenance. See §3.7.

RECORD
Keeps the System of Record: hosts, addresses and DNS names, plus every component, location and project. See Chapter 4.

BUILD
Finds new hosts, enrols them, and installs or reimages machines over the network. See Chapter 5.

RUN THE PLUMBING
Shows the state of Git, the certificate authority and DNS so you can check them in one place. See Chapter 6.

1.2 How the pieces fit

You do not need this section to use the dashboard. It explains where the data comes from, which helps when something looks wrong.

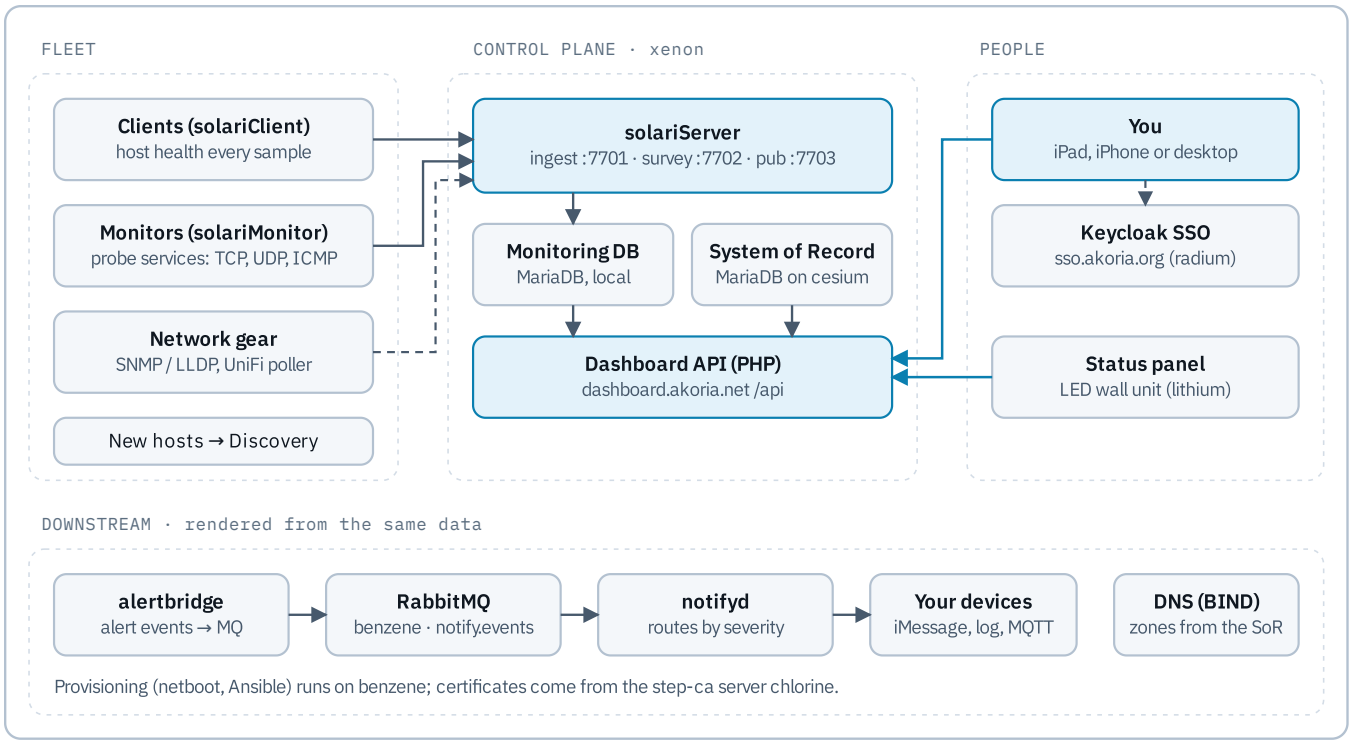


Figure 1.1 How data moves through SolariNet. Solid arrows are live data; dashed arrows are occasional or optional paths. Host names are the machines the services run on today.

- **Agents.** Every enrolled host runs `solariClient`, which reports CPU, memory, disks, network and processes. A few hosts also run `solariMonitor`, which probes services on other hosts. Both connect to `solariServer` on xenon over mutually authenticated TLS, using a certificate issued to each host.
- **Two databases.** The *monitoring database*, on xenon, holds measurements, alerts and rules. The *System of Record* (SoR), on cesium, is the authoritative list of hosts, addresses, DNS names and inventory. Both are MariaDB.
- **Change flows outward.** When the SoR changes, the change is queued on RabbitMQ (on benzene). Services such as the DNS renderer then pick it up, so a host added in the SoR soon has a DNS name.
- **Alerts flow to you.** `alertbridge` turns alert events into messages on RabbitMQ, and `notifyd` delivers them. Separately, the panel daemon on lithium draws health on the LED panel and sounds its alarm.
- **Sign-in** goes through Akoria SSO (Keycloak on radium) or, if enabled, a local account.

1.3 Conventions

YOU SEE	IT MEANS
‹**Save**›	A label exactly as it appears on screen: a button, tab, field or menu item.
3	A numbered callout on the nearby figure.
VIEWER OPERATOR ADMIN	The lowest role that can do the action. Operators can do everything viewers can, and admins everything operators can. See §2.5.
Up Degraded Down Unknown Maintenance	Health colours. The dashboard always shows a word or shape with the colour, and so does this manual.
INFERRED	Worked out from the code without being seen in a running browser. Treat it as likely, not certain.
`/etc/solarinet/…`	A file path on a server. Paths are given so you can find things. Their contents, such as passwords, are never printed here.

> **WIREFRAMES**
> The screen figures are simplified wireframes, not screenshots. They show where things are and what they are called. The live screen has more detail and your own data.

1.4 How to use this manual

- **New to the dashboard:** read Chapter 2, then the section for the screen you need.
- **Looking for one task:** use the search box at the top. It highlights every match and filters the contents list. Press `Enter` to jump between matches.
- **Something is wrong:** go straight to Chapter 7.
- **Unsure of a term:** see the Appendix A.

Each screen section opens with a summary box: what the screen is for, how to reach it, who can change things, and how mature it is. After that come the layout, a table of controls, numbered procedures and notes for iPad and iPhone.

CHAPTER 2

Getting started

Sign in, learn the layout of the dashboard and find your way around. Everything in later chapters builds on this one.

2.1 Signing in

ADDRESS	`https://dashboard.akoria.net` from inside the network or over the VPN
METHODS	Akoria SSO (the usual way) or a local account, if the server has local login turned on
WHO	Anyone with an Akoria directory account. Your role decides what you can change (§2.5).

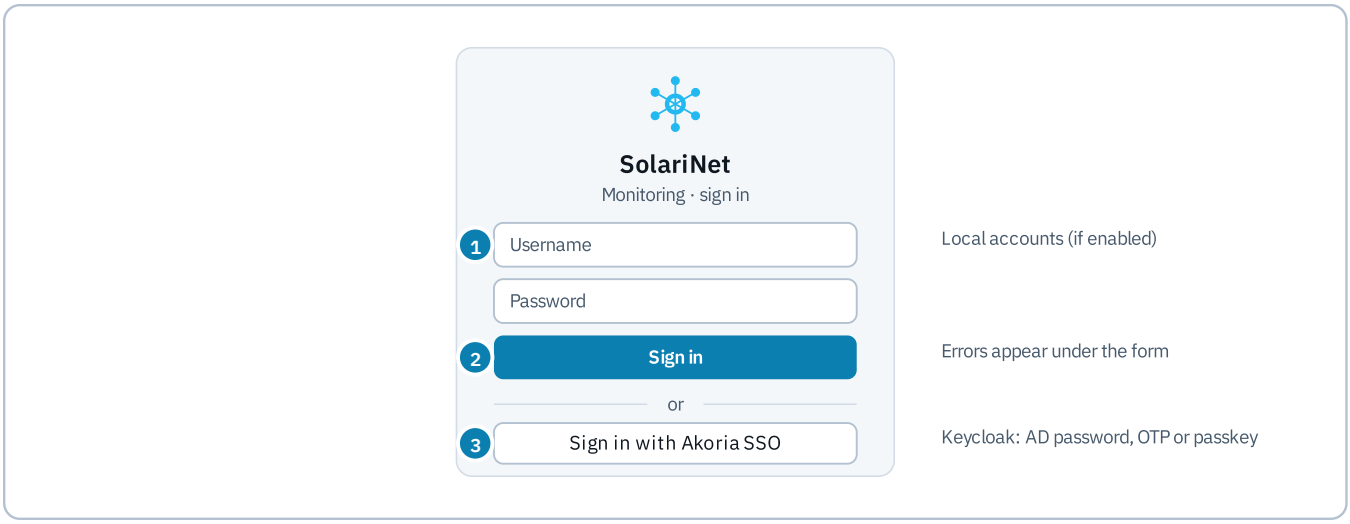


Figure 2.1 The sign-in screen. The local form (1) only appears when local login is enabled on the server. Errors (2) appear above the button. The SSO button (3) appears when SSO is configured; today it reads ‹Sign in with Akoria SSO›.

Sign in with Akoria SSO

1. Open `https://dashboard.akoria.net`. If you are not signed in, the sign-in screen appears.
2. Select ‹**Sign in with Akoria SSO**›. The browser goes to the Akoria sign-in page at `sso.akoria.org`.
3. Sign in there with your Akoria (Active Directory) username, and then your password, one-time code or passkey, whichever the SSO page asks for.
4. The browser comes back to the dashboard and opens ‹**Fleet Overview**›.

The SSO sign-in has to be completed within 10 minutes of selecting the button. If it takes longer, the dashboard shows “Invalid or expired SSO login attempt. Please try again.” Select the button again.

Sign in with a local account

Local accounts are for recovery and service use, for example when SSO is down. They exist only if the server administrator turned local login on.

1. Type the ‹**Username**› and ‹**Password**›.
2. Select ‹**Sign in**›. The button reads ‹**Signing in…**› while it checks.

A wrong username or password shows “Invalid username or password.” The message is the same for both, so it does not reveal which accounts exist. Each username can be tried at most five times from one address in any 15 minutes. After that, sign-in is refused until the oldest attempt is 15 minutes old, even with the right password.

Passkeys and one-time codes

Second factors and passkeys belong to your Akoria SSO account, not to the dashboard. You manage them in the Keycloak account console:

1. Open `https://sso.akoria.org:8443/realms/akoria/account/` and sign in.
2. Open ‹**Account security**›, then ‹**Signing in**›.
3. Under the passkey or authenticator-application heading, choose to set one up and follow the prompts. On an iPad or iPhone, the passkey is saved to iCloud Keychain.

> **NOT VERIFIED**
> The menu names above are Keycloak's standard account-console labels. This manual did not check whether the Akoria realm *requires* a second factor, or which factors it offers. If the account console differs, follow what it shows.

Sessions and signing out

- **Idle timeout:** 30 minutes with no requests ends the session. The dashboard polls every 10 seconds while a tab is open, so an open tab stays signed in.
- **Maximum length:** 8 hours for a local account, 1 hour for an SSO session. After that you sign in again.
- **Role refresh:** for SSO users, the server rechecks your directory groups about every 60 seconds. If that check fails, the session continues as a `VIEWER` until you sign in again.
- **Signing out:** open the profile chip at the top right and select ‹**Log out**›, or press `⌘K` and choose ‹**Log out**›.

> **SHARED DEVICES**
> Logging out of the dashboard probably does not end your Akoria SSO session `INFERRED`. Selecting the SSO button again may sign you straight back in. On a shared device, also sign out of the account console.

2.2 Installing on an iPad or iPhone

The dashboard is a web app that can live on your Home Screen and open full screen, without Safari's toolbars.

1. Open `https://dashboard.akoria.net` in Safari and sign in.
2. Tap the Share button, then ‹**Add to Home Screen**›, then ‹**Add**›.
3. Open SolariNet from the new Home Screen icon. It opens full screen.

> **IPAD AND IPHONE**
> The installed app does not store your data for offline use. Without a connection it shows sample data under a “DEMO DATA” banner (§2.6). It does not send push notifications; alerts reach you through the channels in §3.9.

2.3 The dashboard shell

Every page shares the same frame: a sidebar on the left, a top bar, a banner area, and the page itself.

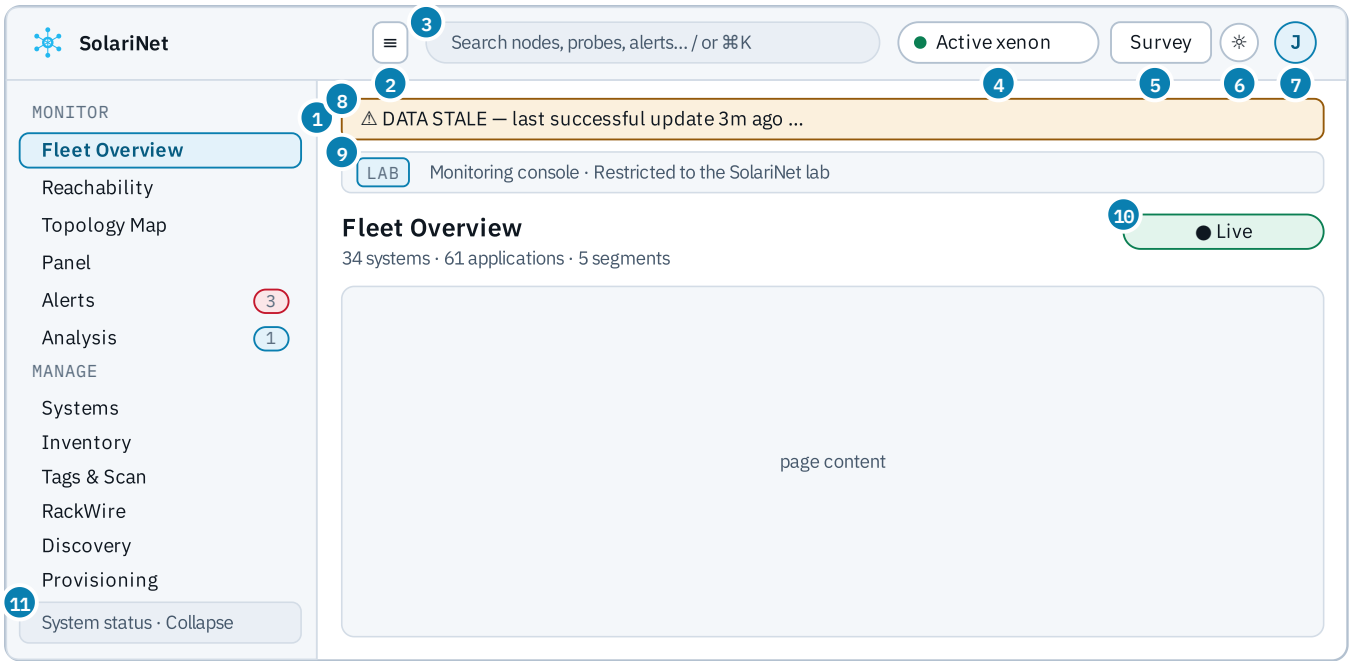


Figure 2.2 The shell. On a screen narrower than 980 px, such as an iPad in portrait, the sidebar is hidden and 2 opens it as a drawer.

#	CONTROL	WHAT IT DOES
1	Sidebar	Navigation in two groups, **Monitor** and **Manage**. Badges show active alert counts (red if any are critical), new Opie reports and new discovery candidates. The sidebar is 248 px wide, or 64 px with icons only when collapsed.
2	Menu button	Collapses or expands the sidebar on a wide screen. On a narrow screen it opens and closes the drawer.
3	Search pill	Opens the command palette (“Command palette”).
4	Server chip	Shows which `solariServer` is active and whether it is reachable.
5	‹**Survey**›	Asks every agent to report now instead of waiting for its next sample. Requires OPERATOR.
6	Theme	Switches between dark and light. The choice is saved in this browser.
7	Profile chip	Shows your initial, name and role. Open it to see the account type and ‹**Log out**›.
8	Data banner	Warns when the data is sample data, stale or incomplete. See §2.6.
9	Maturity banner	States how far the page's feature can be trusted. See §2.7.
10	Live pill	Shown on Fleet Overview. Reads ● **Live** when the data comes from the real API.
11	Status and collapse	A compact fleet summary at the foot of the sidebar, plus the collapse control.

The dashboard reloads its data every 10 seconds. Short confirmation messages ("toasts") appear at the bottom of the screen for about 2.6 seconds.

> ! **KNOWN ISSUE**
> The profile menu shows “Local account” for SSO users too. It only says “Directory account” for a directory source that SSO sessions do not use. The role badge next to your name is correct.

2.4 Finding your way around

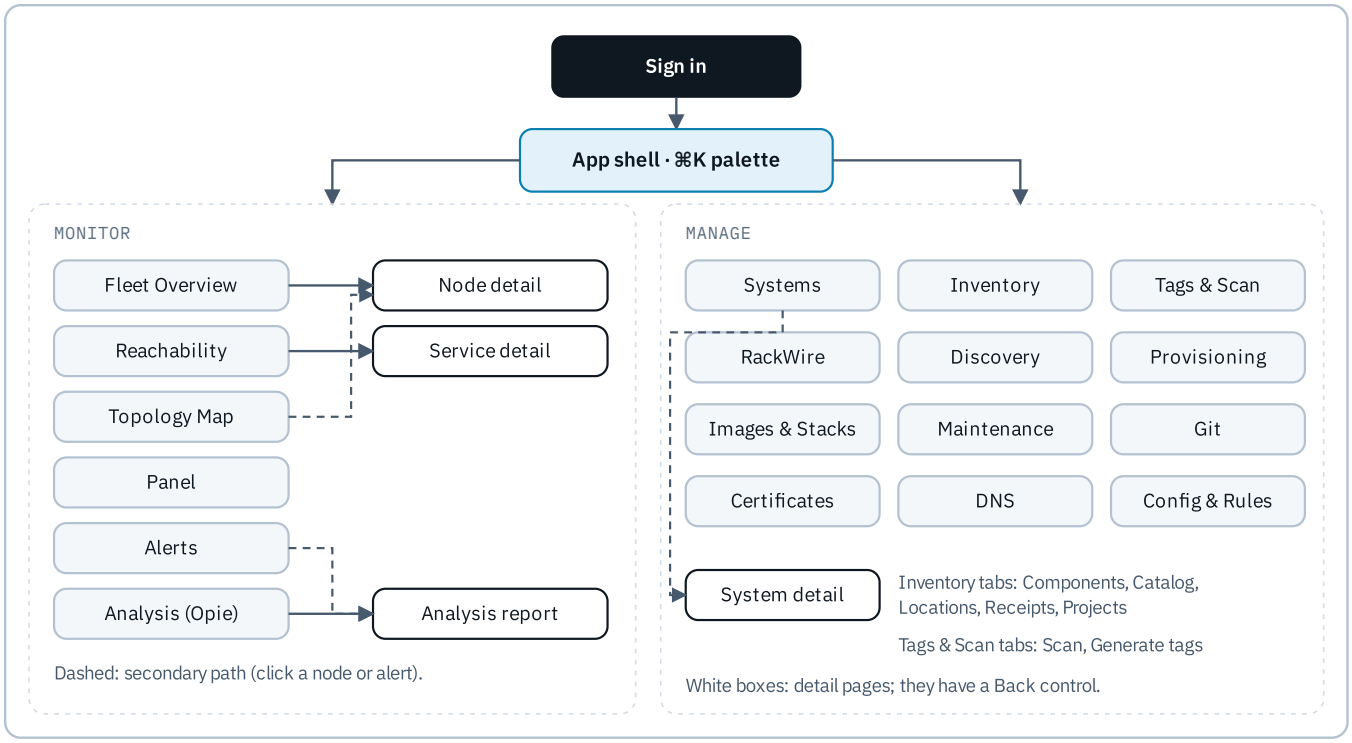


Figure 2.3 Navigation map. Every sidebar entry is a top-level page. Detail pages open from a row, node or report, and return with their Back control or breadcrumb.

SIDEBAR ENTRY	USE IT TO	CHAPTER
Fleet Overview	See the health of every host at a glance	§3.1
Reachability	See which services each monitor can reach	§3.5
Topology Map	See the network and monitoring graph	§3.6
Panel	Watch and control the LED status panel	§3.10
Alerts	Acknowledge alerts; edit alert rules	§3.7
Analysis	Read Opie's automated incident reports	§3.8
Systems	Manage monitored systems, pools and criticality	§3.4
Inventory	Track components, product types, locations, receipts and projects	Chapter 4
Tags & Scan	Scan barcodes; print asset tags	§4.7
RackWire	Plan racks, ports and power	§4.9
Discovery	Find new hosts and start monitoring them	§5.1
Provisioning	Approve addresses, deploy agents, enrol hosts	§5.2
Images & Stacks	Install or reimage machines; apply software	§5.3
Maintenance	Silence alerting during planned work	§5.4
Git	See repositories, commits and pull requests	§6.1
Certificates	Check the CA and host certificates	§6.2
DNS	Query resolvers; check zone parity and records	§6.3
Config & Rules	Change global settings and per-agent configuration	§5.5

Command palette

The command palette is the fastest way to get anywhere.

1. Press `⌘K` (`Ctrl K` on a PC) or `/`, or select the search pill.
2. Type part of a page name, an action or a host name. Results are grouped as **Navigate**, **Actions** and **Jump to node**.
3. Use the arrow keys and `Enter`, or tap a result. `Esc` closes the palette.

The Actions group switches Fleet Overview between Heatmap, Table and Cards, runs ‹**Survey entire fleet now**›, toggles the theme and logs you out. **Jump to node** lists problem hosts first, then up to 40 healthy ones. You can match a host by name, role, IP address or segment.

KNOWN ISSUE

The `/` shortcut also fires while you are typing in some multi-line boxes and drop-down menus. If the palette opens unexpectedly, press `Esc` and keep typing.

2.5 Roles: viewer, operator, admin

Your role is set when you sign in. It does not depend on which device you use.

ROLE	CAN	HOW YOU GET IT
VIEWER	See every page and every record. Cannot change anything.	Default for every SSO user.
OPERATOR	Everything a viewer can, plus every change on every screen. Examples: acknowledge alerts, schedule maintenance, edit inventory, adopt hosts, deploy, reimage, control the panel.	Membership of an operator group named in the server's SSO settings.
ADMIN	Everything an operator can, plus permanent deletion (‹**Purge**›) of a system's tombstone.	Membership of an admin group (by default `Domain Admins`). Local accounts are admins unless configured otherwise.

The server checks every change: by default, signed-in users may read and operators may write. Some screens also hide or disable controls you cannot use. Others show them anyway. If you select one, a message appears: “This action requires an authenticated operator role.” Appendix B lists each action and the role it needs.

QUEUED ACTIONS

Some actions are carried out by the server daemon rather than the web API: survey, push config, re-push, deploy, sign and stage. If the daemon has not answered within about 4 seconds, the dashboard reports the action as **queued**. It will still run. If the daemon cannot be reached at all, you see `control_unavailable` and nothing is queued.

2.6 Data banners

Banners at the top of the page tell you how far to trust what you are looking at.

BANNER	MEANS	WHAT TO DO
● DEMO DATA — live API unreachable	The dashboard could not reach its API when it loaded. It is showing a built-in sample fleet, **not your network**. The reason is shown in brackets.	Check your connection or VPN and reload. If it persists, see §7.2.
● DATA STALE — last successful update Nm ago	The API was working but has stopped answering. The numbers are frozen at the time shown. Changes are blocked until it recovers.	Wait a minute. If you were away for a long time your session may have ended, so reload the page.
● PARTIAL DATA — …	Core fleet data is current, but the listed sections could not be loaded and may be incomplete.	Use the page, but check the named sections later.

> ! KNOWN ISSUE
> The PARTIAL DATA banner does not clear by itself when the missing sections come back. Reload the page to clear it.

2.7 Maturity labels

Each page shows a maturity label. It describes how ready the *feature* is, not whether your network is healthy.

LABEL	MEANING, AS SHOWN	PAGES
Authoritative	Source of truth for the named records.	Inventory, Tags & Scan
Production	Release gates passed for production use.	None yet
Lab	Restricted to the SolariNet lab; not cleared for production use.	Monitoring pages, Maintenance, Git, DNS, Discovery, Provisioning, Images & Stacks, Certificates, Config & Rules, Panel, RackWire
Prototype	Experimental behaviour; treat results as provisional.	Analysis (Opie)
MATURITY UNDECLARED	The page has no maturity entry. This is a defect; report it.	None

Some pages also list related notes under the label. For example, Analysis says automated actions are unavailable, and Provisioning says network-boot installs remain operator-only and lab-only.

2.8 iPad, iPhone and desktop

- **iPad in landscape** (about 1180 px wide) shows the full sidebar. **In portrait** (820 px) the sidebar becomes a drawer: tap the menu button to open it.
- **iPhone** always uses the drawer. Wide tables scroll sideways. Some dense pages, such as Topology and the Reachability matrix, are easier to read on an iPad.
- **Keyboard** shortcuts work with an iPad keyboard. See Appendix C.
- **Barcode scanning** on an iPad works with a Bluetooth or USB scanner that types like a keyboard, or by typing the code. Safari cannot use the camera for this (“Ways to scan”).

CHAPTER 3

Monitoring the network

The Monitor group of the sidebar answers three questions: what is wrong, where, and since when. This chapter covers each Monitor page, plus the Systems pages where you set how much each machine matters.

3.1 Fleet Overview

PURPOSE	See the health of every monitored system at once and jump to whatever needs attention.
OPEN	Sidebar → ‹**Fleet Overview**›. This is the page you land on after signing in.
WHO	`VIEWER` for everything on this page
MATURITY	Lab (Monitoring console)

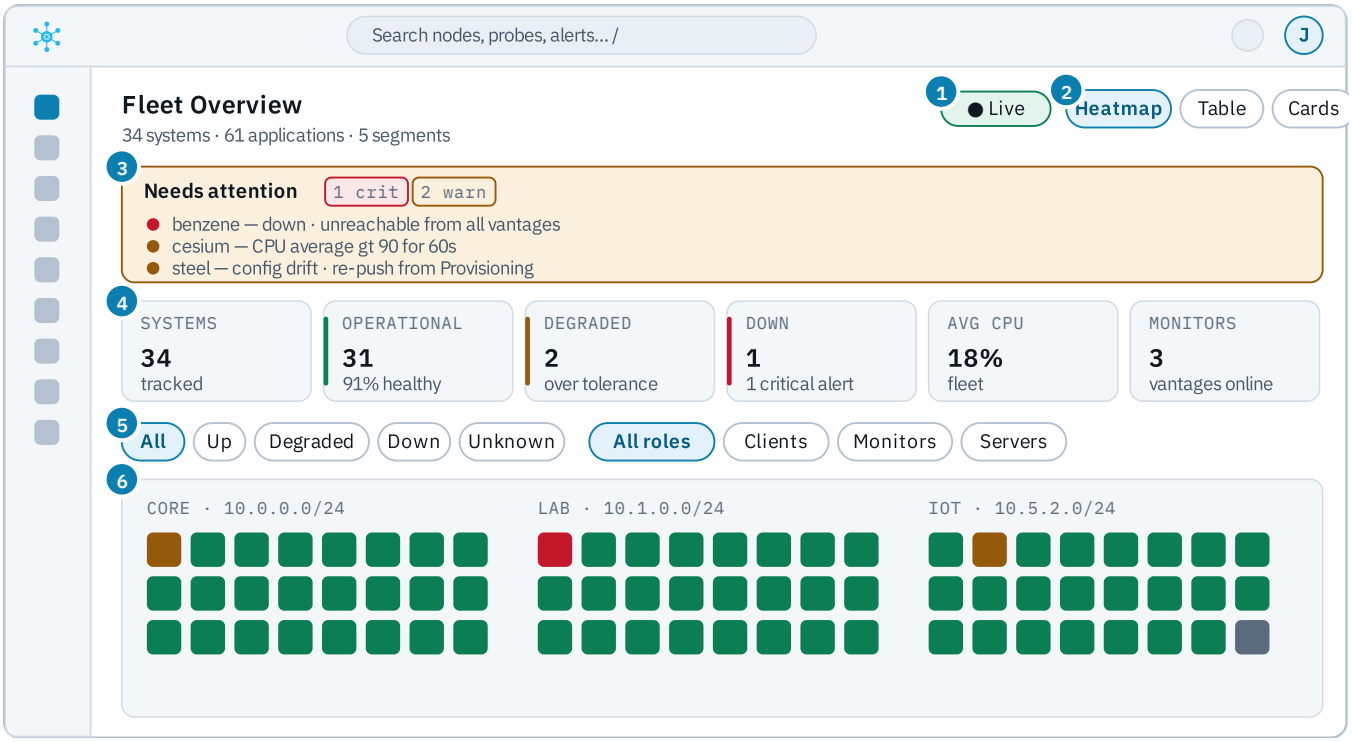


Figure 3.1 Fleet Overview. The pool cards (not drawn) sit above this layout when pools exist.

#	CONTROL	WHAT IT DOES
1	Live pill	Confirms the page shows live data.
2	‹**Heatmap**› · ‹**Table**› · ‹**Cards**›	Three views of the same systems. Heatmap: one tile per system, grouped by network segment. Table: sortable rows. Cards: one card per system with key figures.
3	Needs attention	Problems in priority order: down systems, critical and warning alerts, probe failures, enrolment requests, configuration drift. Counts show as pills. Select a row to go to the cause. Up to seven rows show; ‹**+N more issues**› expands the list. When nothing is wrong, the panel reads “All systems nominal”.
4	Summary tiles	‹**Systems**›, ‹**Operational**›, ‹**Degraded**› and ‹**Down**› also work as filters: select one to show only those systems. ‹**Avg CPU**› and ‹**Monitors**› are information only.
5	Filter chips	State (All, Up, Degraded, Down, Unknown) and role (Clients, Monitors, Servers), each with a count. Two more chips, ‹**Retired**› and ‹**Deleted**›, are the only way to see systems in those states. A density chip packs more tiles per row.
6	Systems	Select a tile, row or card to open its detail page. Hover over a tile (or long-press on iPad) to see its name, state and when it was last seen.

Reading a tile

Tile colour is health: ● **Up**, ● **Degraded** (over a tolerance), ● **Down** (no data or unreachable) and ● **Unknown** (not yet reported). Stale tiles, from hosts that have stopped reporting, are dimmed and their tooltip says “stale”. Critical systems sort first in each segment.

Pool cards

Pools are named groups of systems, such as *DNS* or *Core infra*. When pools exist, a strip of pool cards appears above the fleet view. Each card shows the pool's health. Select a card to open ‹**Systems**›. An `OPERATOR` can delete a pool from its card. Its members move to **Unassigned**, pool 1, which cannot be deleted. To create pools, see §3.4.

> □ IPAD AND IPHONE
>
> Long-press a tile to see its tooltip. On an iPhone the heatmap tiles are small; the ‹**Table**› or ‹**Cards**› view is easier to read.

3.2 Node detail

PURPOSE	Everything about one enrolled host: live gauges, recent trend, interfaces, cores, storage, processes, probes and alerts.
OPEN	Select a host on Fleet Overview, Topology, Reachability or Alerts, or press ⌘K and type its name.
WHO	VIEWER to read; OPERATOR for Survey now, Push config, Retire and criticality

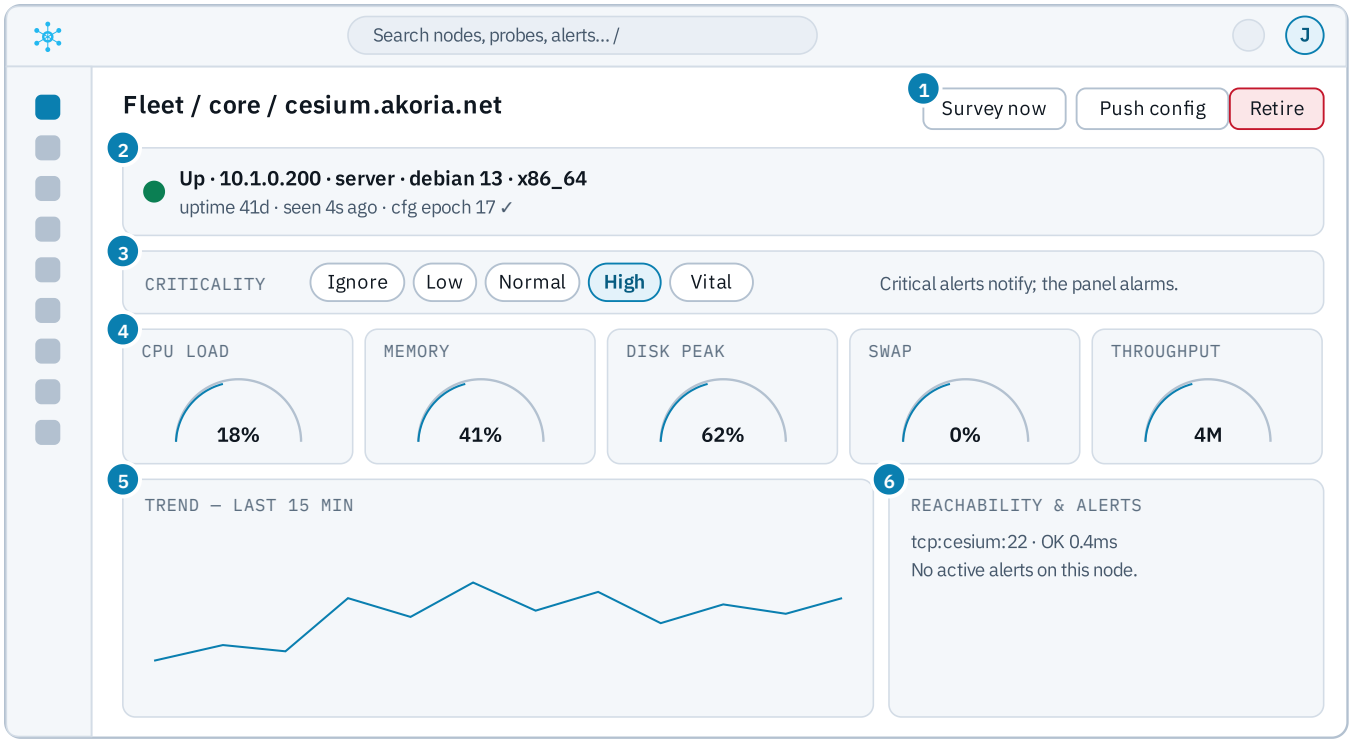


Figure 3.2 Node detail for an agent-reporting host. Systems without an agent open the System detail page instead (“System detail”).

#	AREA	WHAT IT SHOWS OR DOES
1	Actions	‹**Survey now**› asks for a fresh sample. ‹**Push config**› sends the host its configuration again. ‹**Retire**› removes the host from the active roster (see below).
2	Hero	State, IP address, role, operating system, architecture, uptime, when it last reported, and its configuration epoch (a tick means it matches the server).
3	Criticality	How much this host matters. See §3.3.
4	Gauges	CPU load, memory, the fullest disk, swap and network throughput, with their totals.
5	Trend	The last 15 minutes of CPU and memory, plus per-interface traffic, cores, volumes and top processes below.
6	Reachability and alerts	Probes that target this host and their results from each monitor, and any active alerts on it.

Retiring a host

1. On the host's detail page, select ‹**Retire**›.
2. Read the dialog. The host leaves the active roster: no more samples are expected, its open alerts are cleared, and it moves behind the ‹**Retired**› filter. Nothing on the host itself is touched.
3. Select ‹**Retire node**›.

To remove the agent from the machine as well, use ‹**Decommission**› in the agent's configuration dialog instead (“Per-host settings”).

> **KNOWN ISSUE**
> On this page, ‹**Survey now**› and ‹**Push config**› currently survey the *whole fleet*, not just this host. That is harmless, but it is not what the label says.

3.3 Criticality tiers

Criticality tells SolariNet how much a system matters. It changes alert severity, who gets notified and whether the LED panel alarms. Set it on any Node, System or Service detail page with the five-button selector. The behaviour of the selected tier is shown under it as you choose.

TIER	LABEL	BEHAVIOUR, AS THE DASHBOARD STATES IT
0	Ignore	Still monitored and graphed. Never raises an alert.
1	Low	Alerts cap at warning, and nothing is ever notified.
2	Normal	Standard alerting and notification. This is the default.
3	High	Critical alerts notify, and the status panel alarms with them.
4	Vital	Every alert fires critical, notifies at once, and holds the panel alarm while this is down.

Lists show a tier chip only for systems not at Normal, so the chip means "weighted differently". A service inherits its parent system's tier. Changing the tier needs `OPERATOR`. If the server refuses the change, the selector rolls back to the previous tier.

3.4 Systems, pools and lifecycle

PURPOSE	The list of adopted systems (things SolariNet monitors, with or without an agent): rename them, group them into pools, and see their lifecycle state.
OPEN	Sidebar → ‹**Systems**›, or select a pool card
WHO	`VIEWER` to read; `OPERATOR` to rename, re-pool, create pools and change lifecycle; `ADMIN` to purge

The table lists **System**, **IP**, **Class**, **State**, **Targets** (probe count), **Pool** and **Heartbeat**. The heartbeat bars are decorative, not a measurement. Decommissioned systems stay in the list, greyed. Deleted systems are hidden until you select the ‹**Deleted**› chip.

Create a pool and assign systems

1. Select ‹**New pool**› and type a name, for example *DNS*. Select OK.
2. In each system's **Pool** column, choose the new pool from the drop-down menu. The change saves at once.

Rename a system

Select the system's name (it has a small gear), type the new name and select OK. Renaming changes the display name only; the DNS name and SoR host name are separate.

System detail

System detail opens when you select a system that has no agent, for example from Fleet Overview, Alerts or Topology. It shows **Class**, **Pool** and **Heartbeat** tiles, the criticality selector, lifecycle actions and the system's probe targets. Select a target to open Service detail. ‹**Remove**› on a target row stops that one probe, after a confirmation.

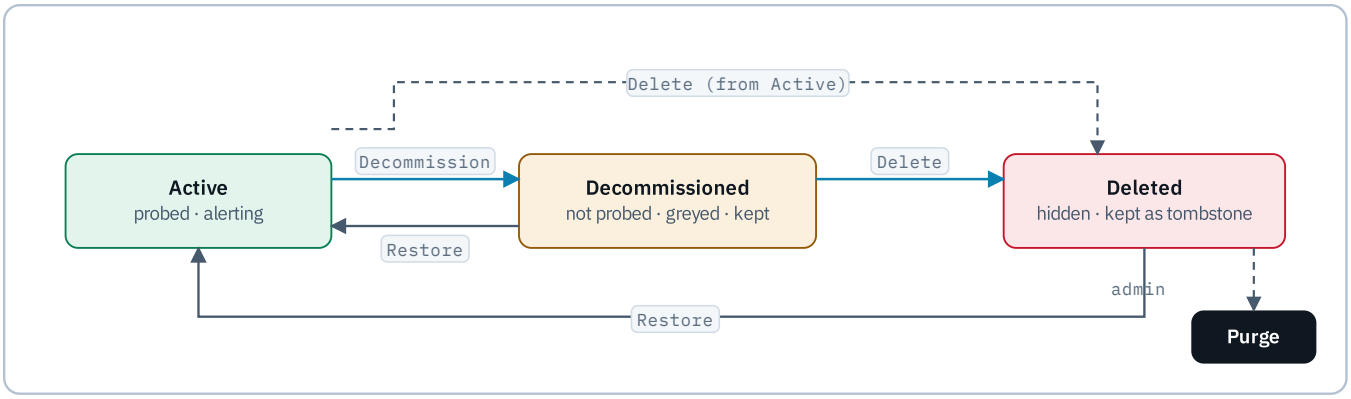


Figure 3.3 System lifecycle. Every state except a purge can be reversed with Restore. Purge is only offered for deleted systems, and only to admins.

ACTION	EFFECT	UNDO	ROLE
‹**Decommission**›	Probing stops, open alerts are cleared, and the system stops counting toward the status panel. It stays in the list, greyed, with its history kept.	‹**Restore**›	OPERATOR
‹**Delete**›	As Decommission, and the system also drops out of the lists. The record is kept so that, if the host turns up on a future scan, Discovery flags it rather than quietly adopting it again.	‹**Restore**›	OPERATOR
‹**Restore**›	Returns the system to active monitoring. Its probe targets are recreated on the next cycle and it alerts again at its current criticality.	—	OPERATOR
‹**Purge**›	Deletes the record and every probe target and reachability sample. Alert history is kept. **This cannot be undone**, and nothing stops the host being adopted again.	None	ADMIN

Purge a deleted system

1. On Systems, select ‹**Deleted**›, then open the system.
2. Select ‹**Purge**›. The dialog shows the stored name to type.
3. Type that name exactly. The server checks it against the stored name.
4. Select ‹**Purge permanently**›. A message confirms “Purged *name* — row and probe history gone”.

Every lifecycle action gives up after 15 seconds with “no response after 15 s — try again”, so a dialog never spins forever.

Service detail

Service detail covers one probe target, such as `tcp:nas:445`. It shows the criticality inherited from the parent system, a **Vantages** table with each monitor's latest result, latency and loss, and ‹**Stop monitoring**› (OPERATOR), which removes the probe.

3.5 Reachability

PURPOSE	Show whether each probed service can be reached from each monitor, so you can tell a dead service from a broken network path.
OPEN	Sidebar → ‹**Reachability**›
WHO	VIEWER; ‹**Survey now**› needs OPERATOR

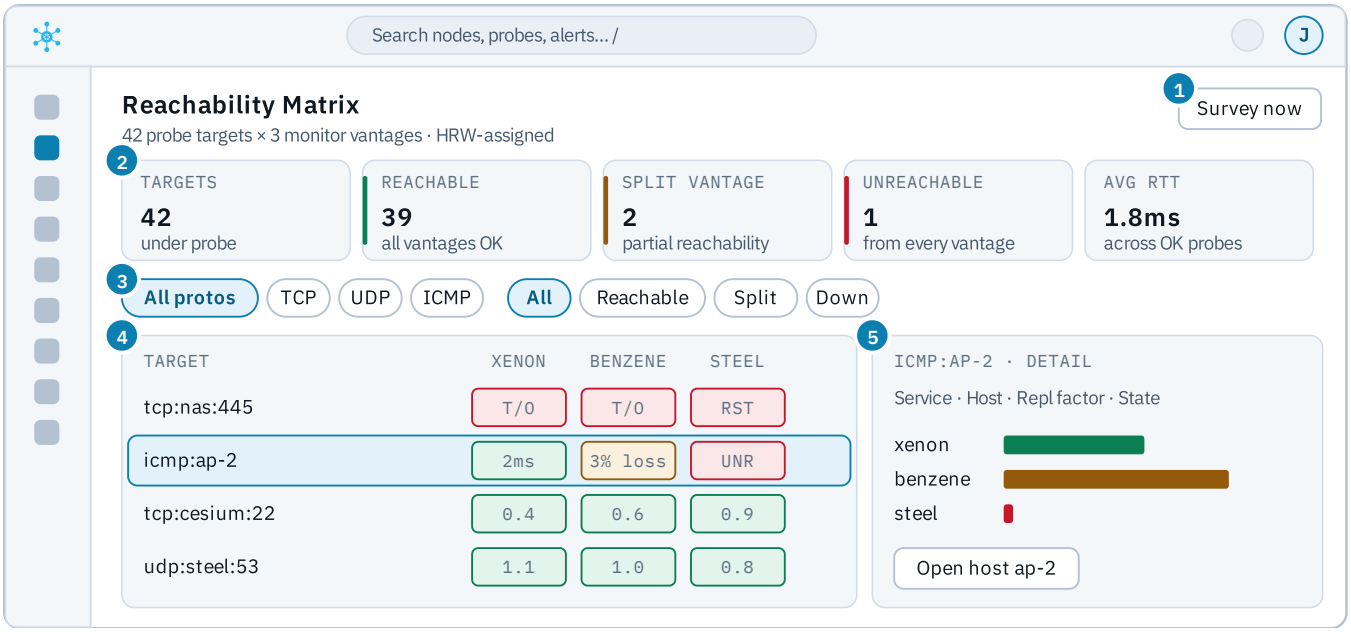


Figure 3.4 The Reachability Matrix. Each row is a probe target and each column a monitor vantage. Each target is assigned to some of the monitors by consistent hashing; this is the replication factor.

#	CONTROL	WHAT IT DOES
1	‹**Survey now**›	Re-runs probes immediately.
2	Summary tiles	‹**Reachable**›, ‹**Split vantage**› and ‹**Unreachable**› filter the matrix. Split vantage means some monitors can reach the target and others cannot, which usually points to a network-path problem.
3	Chips	Filter by protocol (TCP, UDP, ICMP) and by state.
4	Matrix	Each cell shows round-trip time when the probe succeeds, or a code when it fails. Select a row to see its detail.
5	Detail panel	The service, host, replication factor and state, with a bar per vantage, and a link to open the host.

Cell codes

CODE	MEANING	CODE	MEANING
`OK` / a time	Reached; the number is the round-trip time	`DNS`	The target's name did not resolve
`T/O`	Timed out: no answer at all	`TLS`	Connected, but the TLS handshake failed
`RST`	Refused: the host answered, but nothing is listening	`ERR`	Any other probe error
`UNR`	Unreachable: no network route	loss %	Some packets lost (shown amber)

Cells are green when reachable, amber for partial loss and red for a failure, and the code is always shown too.

> **IPAD AND IPHONE**
> The matrix is wide. On an iPhone, turn it sideways or use an iPad.

3.6 Topology Map

PURPOSE	Draw the network (gateways, switches, access points and what hangs off them) or the monitoring graph (which monitors probe what).
OPEN	Sidebar → ‹**Topology Map**›
WHO	`VIEWER`; read-only

A toggle at the top switches between **Network** and **Monitoring**. Network uses LLDP neighbour data read over SNMP and the UniFi poller. Select any node or line to open a card with details. A node's card has ‹**Open node detail**›, and a line's card has ‹**Open endpoint host**›. Small sparklines show recent throughput on network gear.



Figure 3.5 Topology legend. Shape tells you what a node is; colour tells you its health.

LIMITATIONS

The map cannot be panned or zoomed. On an iPhone, turn it sideways or use an iPad. If the topology data fails to load, the page quietly falls back to the sample layout. If the map looks unfamiliar, check for a “DEMO DATA” banner, or compare it with Fleet Overview.

3.7 Alerts and rules

PURPOSE	See and acknowledge what is firing now, review what fired recently, and tune the rules that decide when something is a problem.
OPEN	Sidebar → ‹**Alerts**›. The badge shows the count of active alerts.
WHO	VIEWER to read; OPERATOR to acknowledge and to change rules

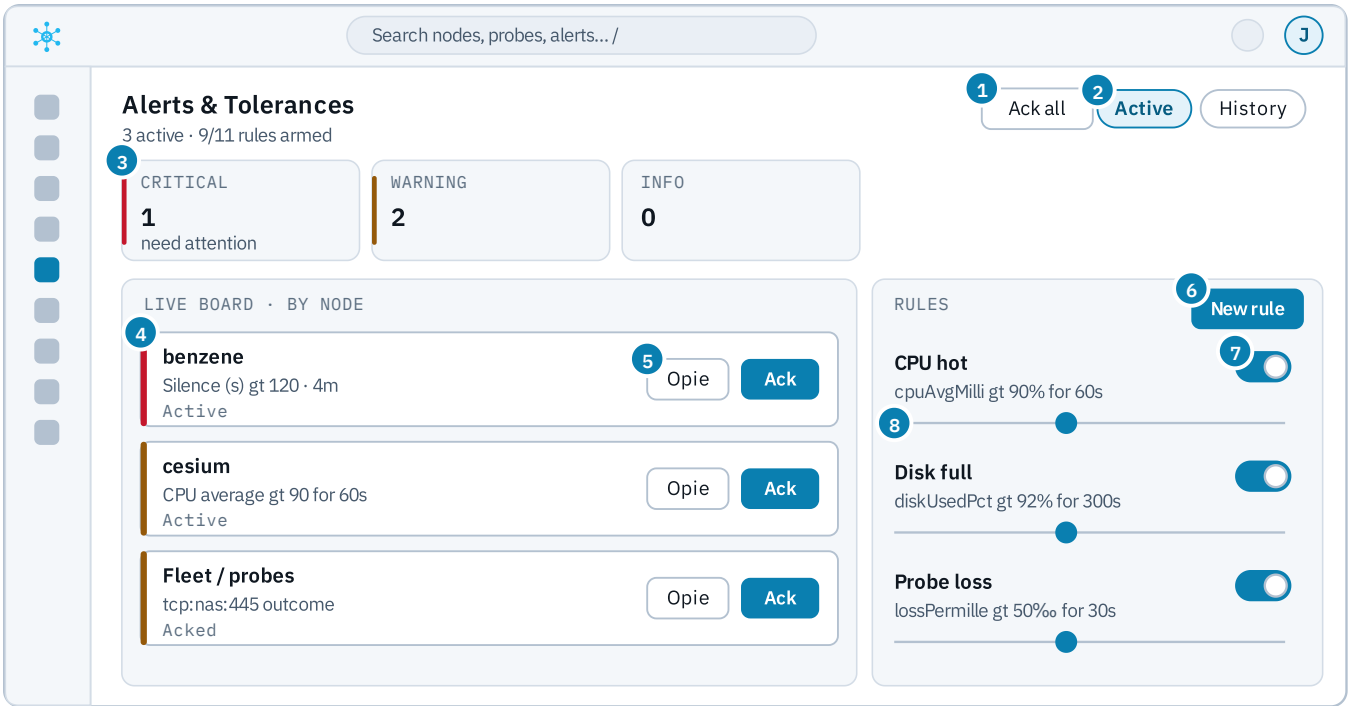


Figure 3.6 Alerts & Tolerances. The live board groups alerts by host; probe alerts with no host are grouped under “Fleet / probes”.

#	CONTROL	WHAT IT DOES
1	‹**Ack all**›	Acknowledges every alert on the Active board, after a confirmation.
2	‹**Active**› · ‹**History**›	Active shows unacknowledged alerts that are firing or cleared within the last 60 minutes. History shows acknowledged alerts and those cleared more than 60 minutes ago.
3	Counts	Critical, Warning and Info alerts still firing and not acknowledged.
4	Alert row	Rule, condition and age. Select the host name to open it.
5	‹**Opie**› · ‹**Ack**›	Opens the Opie report for this incident, if there is one, or acknowledges just this alert.
6	‹**New rule**›	Opens the rule editor.
7	Rule switch	Arms or disarms a rule. The heading counts armed rules, for example “9/11 rules armed”.
8	Threshold slider	Changes a rule's threshold. The new value saves when you release the slider.

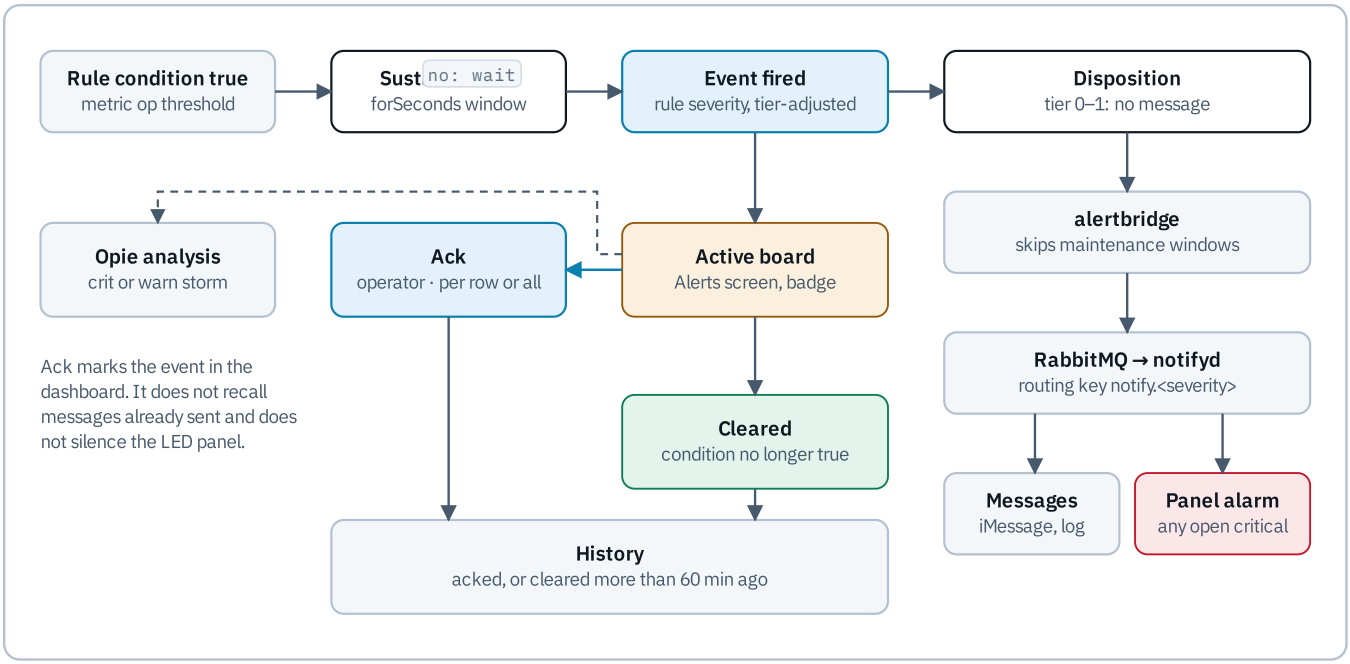


Figure 3.7 The alert lifecycle, from a rule condition to the message on your phone.

Acknowledge alerts

1. On the Active board, find the alert and select ‹**Ack**›. A message confirms “Acknowledged — *rule*” and the row moves to History.
2. To acknowledge everything, select ‹**Ack all**›, then confirm. The message reports how many were acknowledged.

WHAT ACK DOES AND DOES NOT DO

Acknowledging records that someone has seen the alert. It does not fix the problem, recall messages already sent, or silence the LED panel. The panel's alarm has its own acknowledgement (“Acknowledge the alarm”). To stop alerts during planned work, schedule maintenance (§5.4).

Create a rule

1. Select ‹**New rule**›.
2. Type a ‹**Rule name**›, for example *Disk almost full*.
3. Choose the ‹**Scope**›: **host** for agent measurements, or **probe** for service checks.
4. Choose the ‹**Metric**›. Host metrics: CPU average, RAM used, Swap used, Disk used, Silence (last seen), NIC errors, Watched process state. Probe metrics: Packet loss, Round-trip time, Probe outcome.
5. Choose the ‹**Condition**›: greater than, less than, equals, or **on change**. For the first three, set the ‹**Threshold**› (its unit is shown) and ‹**Sustained for (s)**›: how long the condition must hold before the alert fires.
6. Choose the ‹**Severity**›: info, warn or crit. A system's criticality can still change it: Low caps alerts at warning, and Vital raises them to critical.
7. Select ‹**Create rule**›.

KNOWN ISSUES

- Moving the threshold slider with the keyboard changes the number on screen but does not save it. Use a mouse, touch or trackpad and release the slider.
- ‹**Ack all**› also counts alerts that cleared in the last hour, so its total can be larger than the Critical + Warning + Info tiles.

3.8 Analysis (Opie)

PURPOSE	Read automatic first-look reports on incidents.
OPEN	Sidebar → ‹**Analysis**›, or ‹**Opie**› on an alert row
WHO	VIEWER; read-only
MATURITY	Prototype. Treat the findings as a starting point, not a diagnosis.

When a critical alert fires, or a burst of warnings arrives together, Opie gathers recent data about the affected systems. A language model then writes a short report: what happened, the likely cause and what to check. A summary is sent through the notification service.

- The list shows each report's status (**Investigating**, **Done** or **Failed**), its host and its trigger. Filter it with ‹**All**›, ‹**Investigating**› and ‹**Done**›.

- Select a report to read it in full. Select the host name in the report to open the host.
- Opie only reads. It takes no automatic action, and the maturity note on the page says so.

3.9 Notifications

Alert messages are sent by the server, not by the dashboard, so they arrive whether or not a browser is open.

SEVERITY	CONFIGURED CHANNELS ON XENON
crit	log, MQTT, Apple Messages (iMessage), web push, email
warn	log, MQTT, Apple Messages (iMessage), web push, email
info	log, MQTT

The routing table also names SMS and app push for critical alerts, but those senders are not enabled. Nothing is sent when:

- the system is at tier 0 or 1 (Ignore or Low);
- the host is in a maintenance window, based on the server's local time;
- the event is an audit record, not an alert.

The message service also has a dead-man's switch. If the alert feed goes silent for more than 2 minutes (or three sample intervals, whichever is longer), it reports that itself.

> **LIMITS**
>
> - There are no quiet hours and no grouping of duplicate messages. A flapping service can message you each time it changes state. Schedule maintenance, or lower the system's criticality, while you fix it.
> - The dashboard cannot subscribe a browser to web push, so web push reaches no browsers. Do not rely on it.
> - The channels above come from the service's configuration. This manual did not check that each one delivers end to end.

3.10 The LED status panel

A 53 × 11 LED panel on the wall shows fleet health, and it sounds an alarm when something critical needs a person. It is driven by the panel daemon on lithium, which fetches `/api/panel` and sends frames over USB.

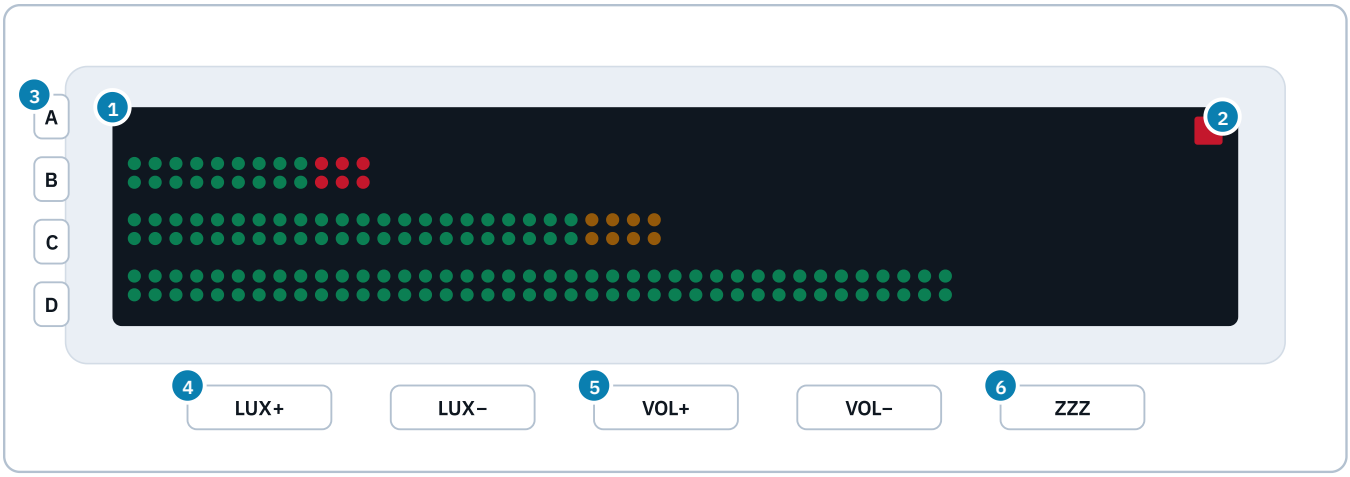


Figure 3.8 The physical panel: the LED matrix 1, alarm beacon 2, theme buttons A–D 3 and the bottom-edge buttons 4–6. Button positions are schematic.

Reading the panel

The panel has four themes, each with three screens. It rotates through them unless you pin one.

THEME	SCREENS
A · Abstract	A0 State lattice · A1 Flow gates · A2 MQ & SNMP
B · Text	B0 Condition counts · B1 Throughput · B2 Load & latency
C · Charts	C0 Load distribution · C1 Live traces · C2 Pool load
D · Instruments	D0 Reachability · D1 Pool bars (the resting and boot screen) · D2 Ambient

Colours always mean the same thing: ● **green** healthy, ● **azure** quiet or information, ● **amber** degraded, ● **red** down, ● **violet** maintenance, and ● **slate** unknown. Brightness shows size or load, never health.

BUTTON	DOES
A, B, C, D	Shows that theme
LUX+ / LUX–	Brightness up or down by 5%. This switches off automatic brightness until you set it back to automatic from the dashboard. Brightness never goes below 25%.
VOL+	Shows the on-panel help
VOL–	Makes the alarm tone quieter. Power-cycling the panel resets it.
ZZZ	Puts the display to sleep
Any button during an alarm	Acknowledges the alarm (below)

The alarm

The alarm goes off when any of these is true:

- a critical alert is open;
- enough of a pool is down that its health score breaks the threshold;
- a **Vital** system is down.

When it goes off, the panel shows an inset naming the problem, lights the beacon and sounds a tone about every 12 seconds. At night the tone is silenced automatically, but the lights still show.

Acknowledge the alarm

1. At the panel, press any button. Or, on the dashboard Panel page, select ‹**Acknowledge**› under **Alarm**.
2. The tone and beacon stop for this *episode*, meaning this particular set of problems. If something new breaks, a new episode starts and the alarm sounds again.

> **THE ALARM TEST IS LOUD**
> Testing the alarm sounds the real tone in the room. Warn the household first.

The Panel page

PURPOSE	See all twelve screens live in the browser, choose which ones rotate, and control the physical panel remotely.
OPEN	Sidebar → ‹**Panel**›
WHO	VIEWER sees everything read-only; OPERATOR uses the controls. If the dashboard cannot confirm your role, the controls stay read-only.

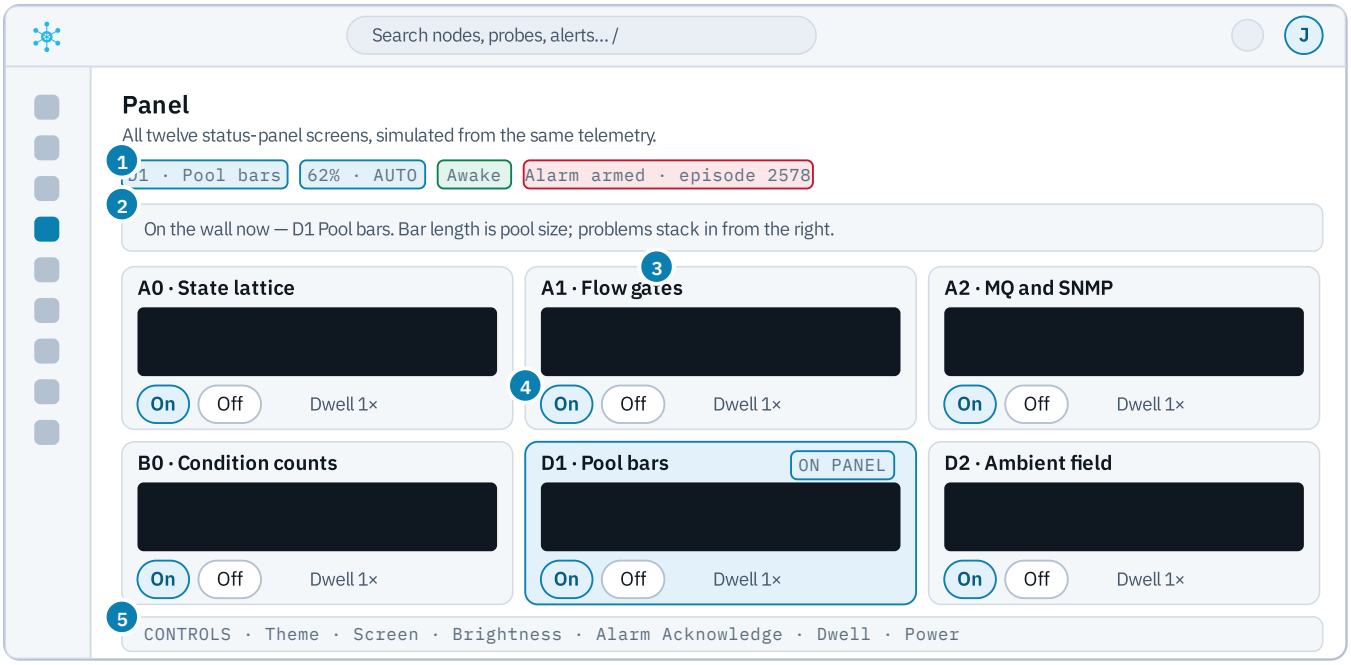


Figure 3.9 The Panel page. Tiles are simulations drawn from the same data the panel receives, refreshed every 5 seconds.

#	AREA	WHAT IT DOES
1	Status chips	The screen showing now, brightness (and whether it is automatic), awake or asleep, and the alarm state with its episode number.
2	On the wall now	Names the screen on the panel and explains how to read it.
3	Screen tile	A live simulation. The tile marked **ON PANEL** is the one showing now.
4	Rotation and dwell	‹**On**›/‹**Off**› includes or skips the screen in rotation. Dwell weight makes it stay up longer.
5	Controls	**Theme**, **Screen**, **Brightness** (or automatic), **Alarm** ‹**Acknowledge**›, **Dwell** (3, 6 or 30 seconds per screen) and **Power** (‹**Wake**› or ‹**Sleep**›).

Each control sends a command that shows as **pending** until the panel confirms it, then **applied**. It shows **failed** with a reason if the panel refuses it. Commands the panel has not collected within 120 seconds expire.

PANEL SHOWS	MEANS	DO
NO DATA	The daemon is running but cannot get fresh data from the dashboard API	Check the dashboard and server; the panel recovers by itself
LINK LOST	The panel is not hearing from the daemon on lithium	Reseat the panel's USB cable, or restart the `solari-panel` service on lithium

CHAPTER 4

Inventory, tags and RackWire

Inventory is the record of what Akoria owns and where it is. It lives in the System of Record on cesium, is the authoritative copy, and is shared with RackWire and provisioning. This chapter covers the Inventory screen, barcode tags and scanning, labels, and the RackWire planner.

4.1 Concepts

Read this section once; every inventory screen uses these terms.

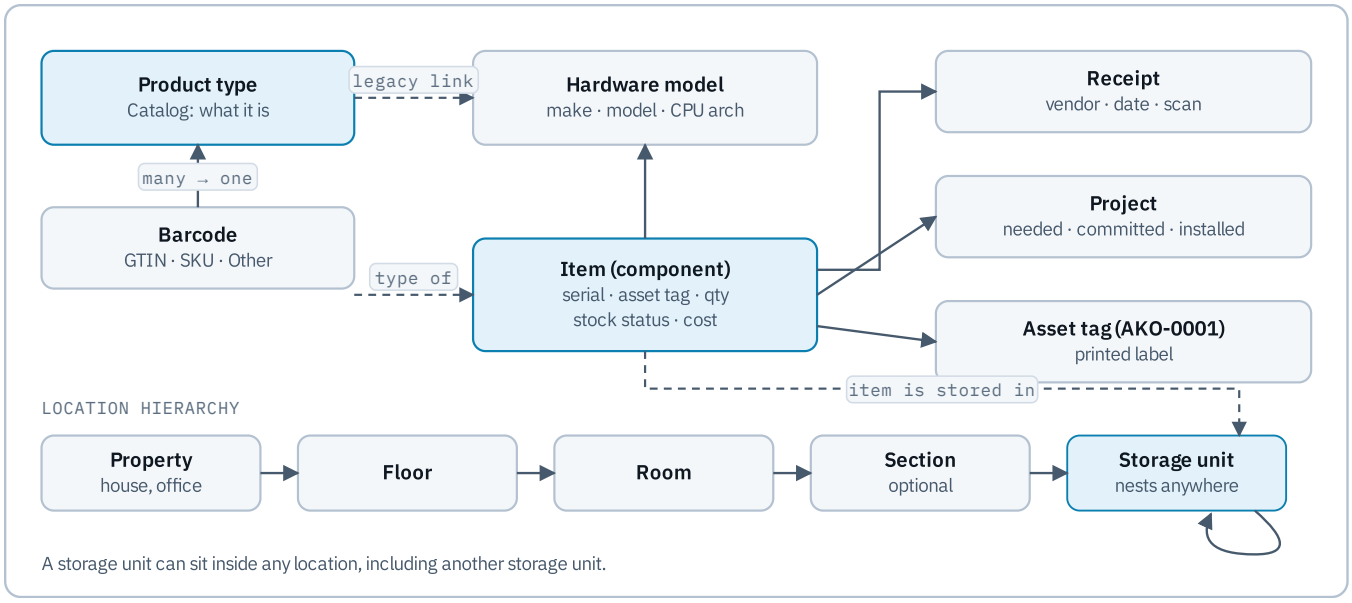


Figure 4.1 How inventory records relate. A catalog type says *what* something is; an item is one physical thing you own; a location says *where* it is.

TERM	MEANING
Product type (Catalog)	What something is: a make and model, a software title, a consumable. It has a kind (hardware, peripheral, component, software, accessory, consumable, other), optional specifications and any number of barcodes. Owning three of the same thing means one product type and three items.
Barcode	A code printed on the product or its box (GTIN/UPC/EAN, a shop SKU or anything else) attached to a product type. Each code belongs to one product type, so scanning it finds the type.
Hardware model	The make, model, device type and CPU architecture used by provisioning and RackWire. Hardware product types are linked to a hardware model.
Item (component)	One thing you own, or a batch of identical things: serial, asset tag, quantity, stock status, cost, location, receipt. The Components tab lists them.
Stock status	**in stock** (available), **allocated** (committed to a project), **deployed** (installed and in use), **consumed**, **retired**.
Location	Where something is. Locations form a tree: **property** (a house or site) → **floor** → **room** → **section** (optional). A **storage unit** (drawer, bin, shelf, cabinet) can sit inside any location, including another storage unit.
Project	A build that reserves items: what is needed, what is committed, what is installed, and what still has to be bought. A project can have a **tray**, a storage unit where its parts are gathered.
Tag	An Akoria label code: `AKO-0142` for an item, `LOC-0031` for a location, `PRJ-0004` for a project. Printed as a QR code and a Code 128 barcode.
SoR entity	The network record (host, device or service) in the System of Record. An item can be linked to the entity it physically is, so a server's box and its network identity are tied together.

WHO CAN CHANGE INVENTORY

Everyone signed in can read inventory. Changes need `OPERATOR`. The inventory screens show every control to everyone; if you are a viewer, a change fails with a message such as “Save failed: This action requires an authenticated operator role.” Scanning also needs `OPERATOR`, because every scan is recorded.

LIVE CONNECTION NEEDED

Changes are refused while the dashboard is not connected live (§2.6). The Catalog, location changes, Tags & Scan and RackWire profiles are unavailable offline; the Components list stays readable.

4.2 Components

PURPOSE	Browse and edit every item you own.
OPEN	Sidebar → ‹**Inventory**›. Components is the first tab.
WHO	VIEWER to read; OPERATOR to add, edit and link
MATURITY	Authoritative: this is the source of truth for inventory.

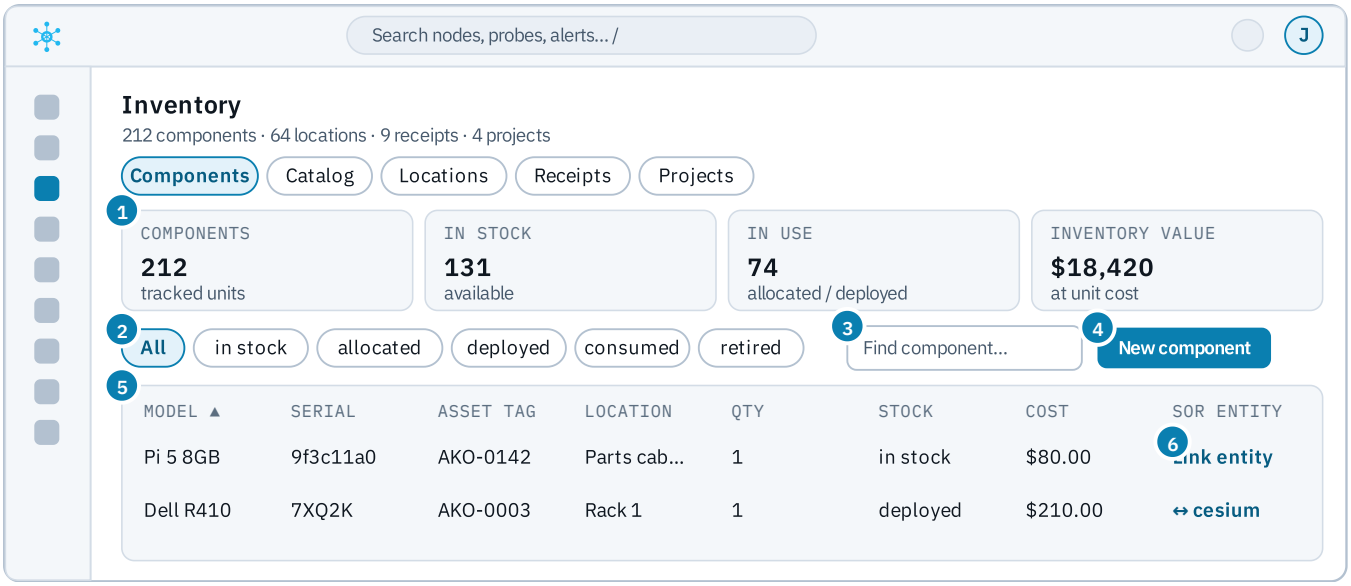


Figure 4.2 The Components tab. The page heading counts components, locations, receipts and projects.

#	CONTROL	WHAT IT DOES
1	Summary tiles	**Components** (tracked units), **In stock**, **In use** (allocated or deployed) and **Inventory value** at unit cost.
2	Stock chips	Show all items or only one stock status. Each chip shows a count.
3	Find component	Matches model, serial, asset tag and location.
4	‹**New component**›	Opens the component form. ‹**RackWire device profiles**›, beside it, opens the profile editor (“RackWire device profiles”).
5	Table	Model, Serial, Asset tag, Location, Qty, Stock and Cost sort when you select the heading. Select a row to edit the item. Select the receipt to view it. 100 rows per page.
6	SoR entity	‹**Link entity**›, or ‹**↔ *name***› when linked, opens the entity link panel below the table.

Add or edit a component

1. Select ‹**New component**›, or select an existing row.

2. Choose the ‹**Model**›. If it is not listed, select ‹**+**› beside it to add a make and model, its device type and its ‹**CPU architecture**›. Provisioning will only reimage a machine whose architecture is known.
3. Set ‹**Stock status**›, ‹**Serial**›, ‹**Asset tag**›, ‹**Location**›, ‹**Quantity**›, ‹**Acquired on**›, ‹**Unit cost (USD)**›, the ‹**Receipt**› and any ‹**Notes**›.
4. Select ‹**Add component**› or ‹**Save changes**›.

The model list does not offer retired catalog types. An item does not need a hardware model at all: software, consumables and other non-hardware items are recorded against their catalog type. The quickest way to add items of a known type is ‹**Add item(s)**› on the Catalog (“Record items of a type”).

Link an item to its SoR entity

Linking says "this physical box *is* that network host". RackWire uses the link when it resolves an asset tag for cable planning. Names are shown as candidates but never matched automatically.

1. In the item's **SoR entity** cell, select ‹**Link entity**›.
2. In the panel **Physical identity ↔ SoR entity**, type part of a host name or role in ‹**Find a SoR entity**› and select ‹**Search**›.
3. Choose the entity and select ‹**Link entity**›. The panel confirms “Entity association saved.”

To undo, open the panel again and select ‹**Remove association**›. If the entity does not exist yet, create the host record first (for example by adopting it in Discovery).

RackWire device profiles

A profile describes a hardware model's ports and power so RackWire can place it with correct port geometry. Select ‹**RackWire device profiles**›, choose the ‹**SoR hardware model**› (a tick marks models that already have a profile), then fill in:

- ‹**Planner label**›, ‹**Device kind**›, ‹**Typical draw (W)**› and ‹**PoE output budget (W)**›.
- One or more **connection groups**, each with a label, connector (RJ45, SFP, USB-A, USB-C, IEC in, IEC out, NEMA, barrel, HDMI), count (1–48), speed in Gbps, watts, and a PoE standard with watts per port.

Select ‹**Save RackWire profile**›. The model then appears in RackWire's gear library. Check port counts and power against the device's specification; RackWire's compatibility checks rely on these numbers.

> **IPAD AND IPHONE**
> The table is wider than an iPad in portrait and scrolls sideways; the SoR entity and Receipt columns are at the right. Forms open as sheets over the page.

4.3 Catalog

PURPOSE	Keep one product type for each kind of thing you own, with its barcodes, so a scan identifies it.
OPEN	Sidebar → ‹**Inventory**› → ‹**Catalog**› tab
WHO	VIEWER to read and look up; OPERATOR to add, edit, attach codes and retire

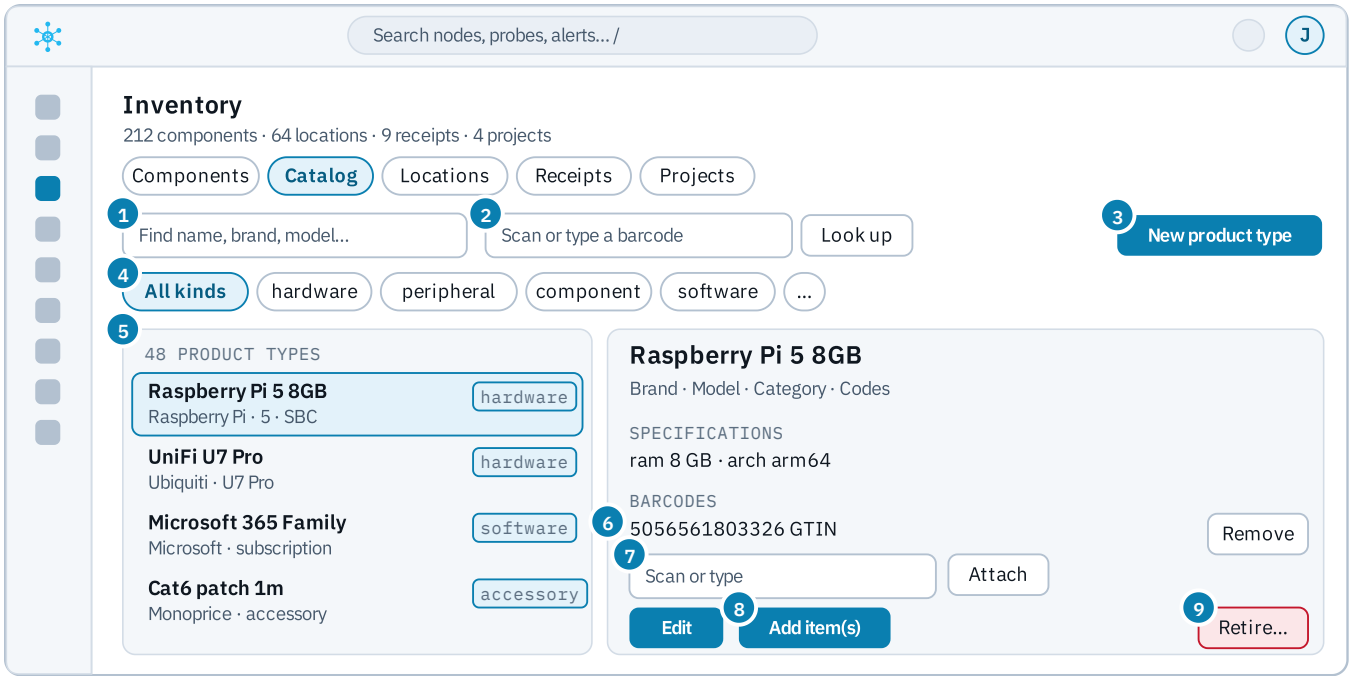


Figure 4.3 The Catalog tab: the product type list on the left and the selected type's details on the right.

#	CONTROL	WHAT IT DOES
1	Find	Searches name, brand, model and category as you type.
2	Barcode lookup	Scan or type a code and select ‹**Look up**›. Finds the product type that owns the code. It never creates anything.
3	‹**New product type**›	Opens the product type form.
4	Kind chips	Filter by kind.
5	List	Name, brand · model · category and kind. 50 at a time; ‹**Show more**› loads the next 50.
6	Barcodes	Codes attached to this type. ‹**Remove**› detaches one after a confirmation.
7	Attach a code	Scan or type a code, choose its ‹**Code type**› and select ‹**Attach**›.
8	‹**Edit**› · ‹**Add item(s)**›	Edit the type, or record items of this type (“Record items of a type”).
9	‹**Retire…**›	Retires the type (below).

Look up a barcode

1. Put the cursor in ‹**Scan or type a barcode**› and scan the code. The scanner's Enter does not submit the form, so nothing happens by accident.
2. Select ‹**Look up**›.
3. If the code is known, the page says “*code* is *name*.” and opens that type.
4. If not, it says “No product type has the code … Nothing has been saved.” Choose one:
   - ‹**New product type with this code**› opens the form with the code filled in;
   - ‹**Use for *type***› attaches the code to the type you have open;
   - ‹**Dismiss**›.

Lookup only checks Akoria's own catalog. It does not search the internet, so a new product needs its details typed once.

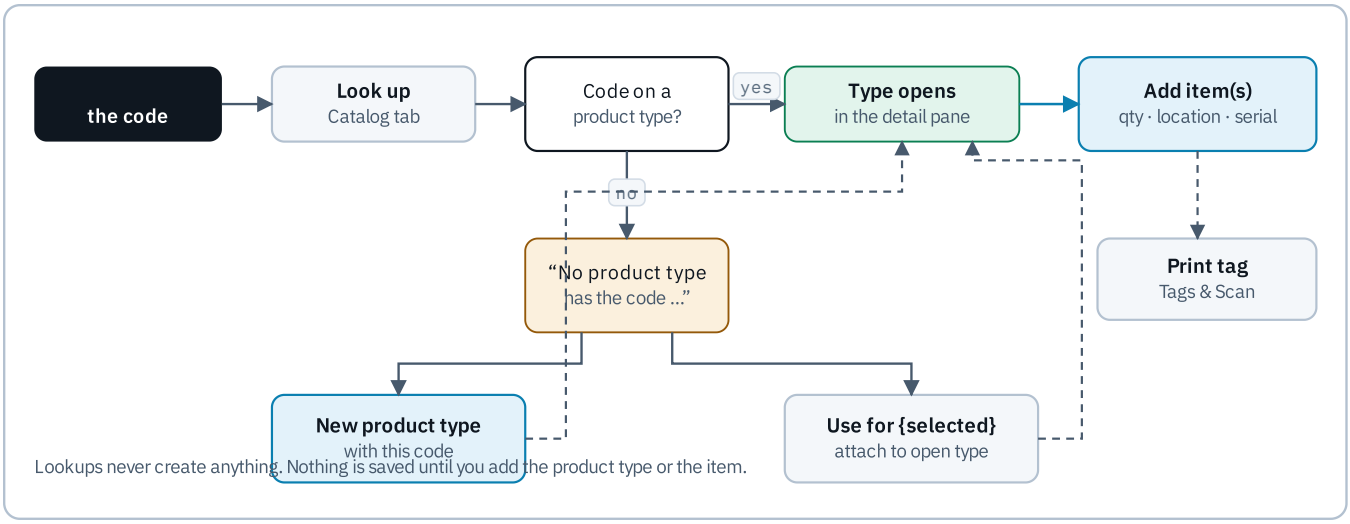


Figure 4.4 From a barcode to recorded items. The scan only looks up; nothing is saved until you add the product type or the items.

Add a product type

1. Select ‹**New product type**›.
2. Fill in ‹**Brand**› and ‹**Model**›. The ‹**Display name**› is built from them unless you type your own (the form shows the name it will use).
3. Choose the ‹**Kind**› and, optionally, a ‹**Category**›, such as *Single-board computer*.
4. Add **Specifications** as name and value pairs, for example *ram = 4 GB*. Select ‹**Add spec**› for each.
5. Optionally add an ‹**Image link**› (an http or https address), ‹**Notes**› and a first ‹**Barcode**› with its ‹**Code type**›.
6. Select ‹**Add product type**›.

Code types

TYPE	USE FOR	RULE
GTIN / UPC / EAN	The retail barcode on the box	8, 12, 13 or 14 digits with a correct check digit. A wrong digit is refused, which catches mis-scans.
SKU	A shop's own code	Up to 128 characters
Other	Anything else	Up to 128 characters

NO SECRETS IN THE CATALOG

Specification names that look like secrets, such as *licence key*, *password* or *token*, are refused. Licence keys are never stored in SolariNet. For software, record a *reference* to where the licence is kept, on each item.

Record items of a type

1. Open the product type in the Catalog and select ‹**Add item(s)**›.
2. Enter the **quantity** and choose the **location**.
3. For a single item, add its **serial** and **asset tag** if it has them.
4. For **software**, record the number of licence **seats**, the **expiry** date, a **licence reference** (where the licence is kept, never the key itself) and, optionally, the hardware item it is **installed on**. "Installed on" must be a physical item, not another piece of software.
5. Save. The new items appear on the Components tab.

Retire a product type

Select ‹**Retire…**›, then ‹**Retire product type**›. Items already recorded keep the type, but it leaves the pick lists and its barcodes are released so they can be attached elsewhere. You can retire a type that is still in use.

> **LINKED TYPES**
> A type marked “Linked to a component model” is the same record as a hardware model. Changing its brand, model, notes or kind also changes that model in Components, provisioning and RackWire.

4.4 Locations

PURPOSE	Build and maintain the tree of places and storage units.
OPEN	Sidebar → ‹**Inventory**› → ‹**Locations**› tab
WHO	VIEWER to browse; OPERATOR to change

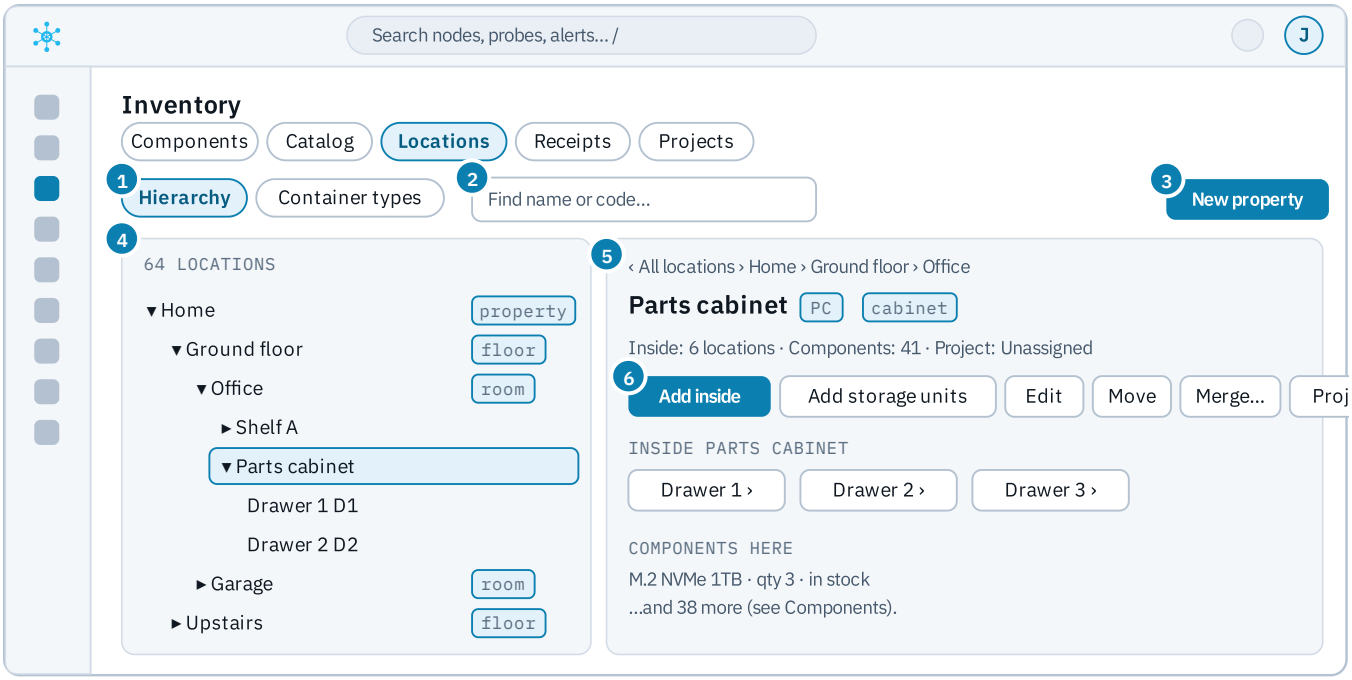


Figure 4.5 The Locations tab. The tree is on the left; the selected location's path, contents and actions are on the right.

#	CONTROL	WHAT IT DOES
1	‹**Hierarchy**› · ‹**Container types**›	Switch between the location tree and the list of container types (drawer, bin, tray, shelf...).
2	Find	Finds locations by name or code, anywhere in the tree, and shows their full path. Select a result to open it.
3	‹**New property**›	Adds a top-level location: a house, office or site.
4	Tree	Select ▸ or ▾ to open or close a branch. Each row shows its code, level and item count.
5	Detail	Breadcrumb path, level, code, container type and project, then what is inside it and the components stored there.
6	Actions	‹**Add inside**›, ‹**Add storage units**›, ‹**Edit**›, ‹**Move**›, ‹**Merge...**› and, on storage units, ‹**Project**›.

What can go inside what

LEVEL	GOES INSIDE
Property	Nothing: properties are always top level
Floor	A property
Room	A floor
Section	A room (optional level)
Storage unit	Anything, including another storage unit

Add a location

1. Select the parent in the tree, then ‹**Add inside**›. For a new house or site, use ‹**New property**› instead.
2. Type a ‹**Name**› (up to 128 characters) and, optionally, a short ‹**Code**› such as *D3* (up to 32 characters).
3. Choose the ‹**Level**› if more than one is allowed here. For a storage unit, optionally choose a ‹**Container type**› and a ‹**Project**›.
4. Select ‹**Add location**›.

Codes are unique across all live locations, so the same code cannot be used in two places. Names only need to be unique within the same parent.

Add many storage units at once

1. Select the parent and then ‹**Add storage units**›.
2. Choose ‹**Numbered**› and give ‹**How many**› (1–500), a ‹**Name prefix**› such as *Bin-* and a ‹**Start at**› number. Or choose ‹**Paste names**› and paste one name per line.
3. Optionally give a ‹**Code prefix**› and ‹**Code start**›, a container type and a project.
4. Check the **Preview**; any clash is listed above it.
5. Select ‹**Create *N* storage units**›. It is all-or-none: if any name or code clashes, nothing is created.

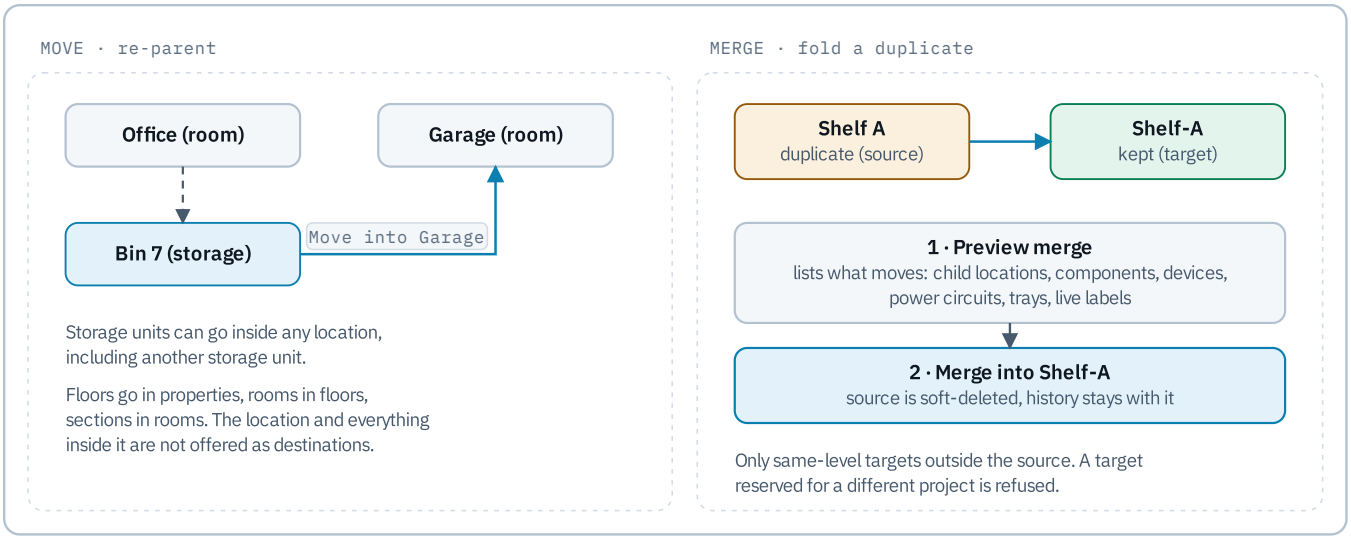


Figure 4.6 Moving and merging locations.

Move a location

1. Select the location and then ‹**Move**›. (Properties cannot be moved.)
2. Choose a destination. Only legal destinations are listed. The location itself and everything inside it are left out, so you cannot create a loop.
3. Select ‹**Move into *destination***›. Everything inside moves with it.

Merge duplicate locations

Use Merge when the same place was entered twice, for example *Shelf A* and *Shelf-A*.

1. Select the duplicate (the one to get rid of) and then ‹**Merge…**›. The button only appears if another location of the same level exists.
2. Choose the location to keep. Only same-level locations outside the duplicate are offered. A location reserved for a different project is shown but cannot be chosen.
3. Select ‹**Preview merge**›. The preview lists what moves to the kept location (child locations, components, devices, power circuits, project trays, live barcode labels) and what stays with the duplicate (scan history and external references).
4. Select ‹**Merge into *kept***›.

The duplicate is soft-deleted: hidden, but still in the audit trail with its history. A merge cannot be undone from the dashboard. If something changed after the preview, the merge is refused with “Merge refused: … Preview the merge again before confirming.”

Reserve a storage unit for a project

On a storage unit, select ‹**Project**›, choose the project and select ‹**Assign**›. To release it, select ‹**Release**›; the container must be empty first. Changing a storage unit's level also releases it, and has the same rule.

Container types

Under ‹**Container types**›, add a type with ‹**Add type**›, or ‹**Rename**› or ‹**Retire**› an existing one. Retiring keeps the type on containers that already have it but removes it from pick lists. ‹**Include retired**› shows retired types.

If someone else changed it first

Edits check that the record has not changed since you opened it. If it has, you see “This was changed elsewhere since you opened it. The latest version has been loaded — check it and save again.” Review the reloaded values and save again.

This check is precise to the microsecond, so two quick saves in the same second are still told apart. If the database is momentarily busy with a conflicting change, the save is refused with a 409 asking you to **try again**; nothing was changed, so simply repeat it.

> □ IPAD AND IPHONE
> The Catalog and Locations tabs use two panes, list and detail. On a narrow screen, scroll down from the list to reach the detail pane INFERRED. Use an iPad in landscape for long location work.

4.5 Receipts

PURPOSE	Keep purchase records and their scans, and see which items came on each.
OPEN	Sidebar → ‹**Inventory**› → ‹**Receipts**› tab
WHO	VIEWER to read; OPERATOR to upload

The tab shows the number of receipts and the total spend, then one card per receipt, newest first. Select a card to see the scan, notes and linked components.

1. Select ‹**Upload receipt**›.
2. Fill in ‹**Vendor**›, ‹**Purchased on**›, ‹**Subtotal (USD)**› and ‹**Order ref**›.
3. Under ‹**Scan (image or PDF)**›, choose the file. On iPad you can take a photo or pick from Files.
4. Select ‹**Save receipt**›. Then link items to it from each component's ‹**Receipt**› field.

> ! ALWAYS ATTACH A SCAN
> A receipt saved *without* a scan file shows “Receipt saved” but is not stored. Until this is fixed, attach a scan or photo every time. If you have none, photograph the order confirmation.

4.6 Projects

PURPOSE	Reserve owned items for a build, track installation and list what still has to be bought.
OPEN	Sidebar → ‹**Inventory**› → ‹**Projects**› tab
WHO	VIEWER to read; OPERATOR to change

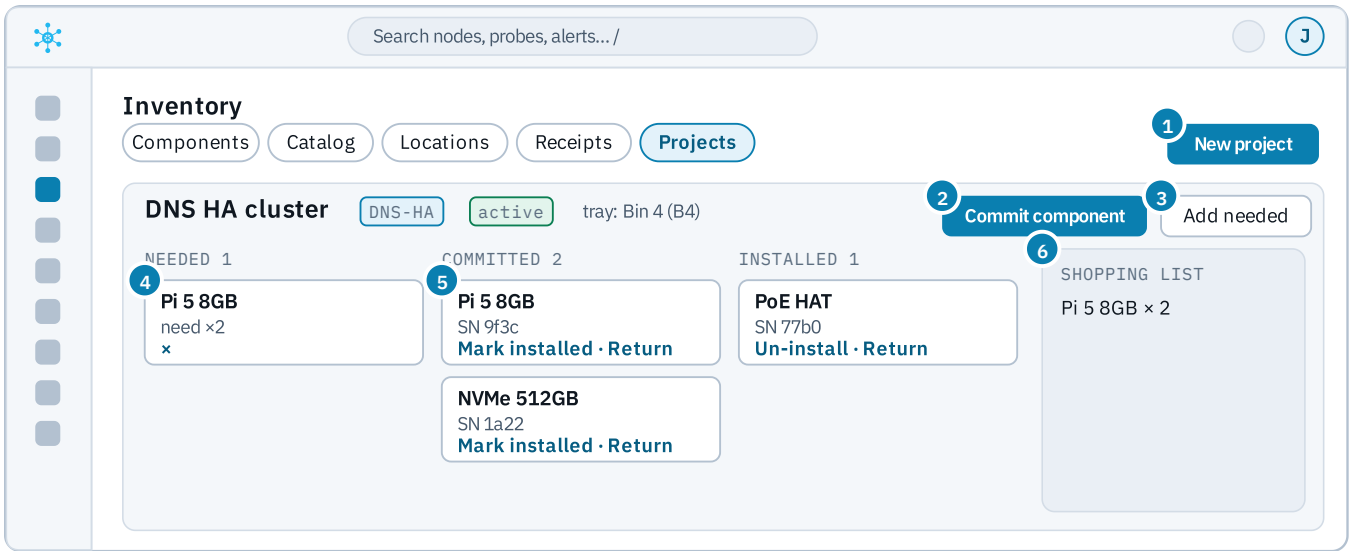


Figure 4.7 A project board. Items move from Needed to Committed to Installed.

#	CONTROL	WHAT IT DOES
1	‹**New project**›	Name, code, status, tray, owner, target date and description.
2	‹**Commit component**›	Reserves an item you own for this project and moves it to the project's tray.
3	‹**Add needed model**›	Records something the project needs that you do not own yet.
4	Needed	Things required but not yet committed. ‹**×**› removes one.
5	Committed · Installed	Committed items show ‹**Mark installed**› and ‹**Return**›. Installed items show ‹**Un-install**› and ‹**Return**›.
6	Shopping list	Needed models with no item committed, with quantities. When empty: “Nothing to buy”.

What each step does to the item

ACTION	ITEM'S STOCK STATUS BECOMES
Commit	allocated, and the item moves to the project's tray
Mark installed (or commit as installed)	deployed
Return	in stock

An item can be committed to only one project at a time. Trying to commit it to a second project fails with “409 · unit already committed to another project”; return it from the first project first. Project details cannot be edited after creation; the status and tray are set when you create the project.

4.7 Tags & Scan

PURPOSE Scan any code to identify it, move items into bins, commit items to projects, receive new stock, and print labels.

OPEN Sidebar → ‹**Tags & Scan**›

WHO OPERATOR. Every scan is looked up and written to the audit log.

NEEDS The live API. Offline, the page says “Barcode tags need the live API”.

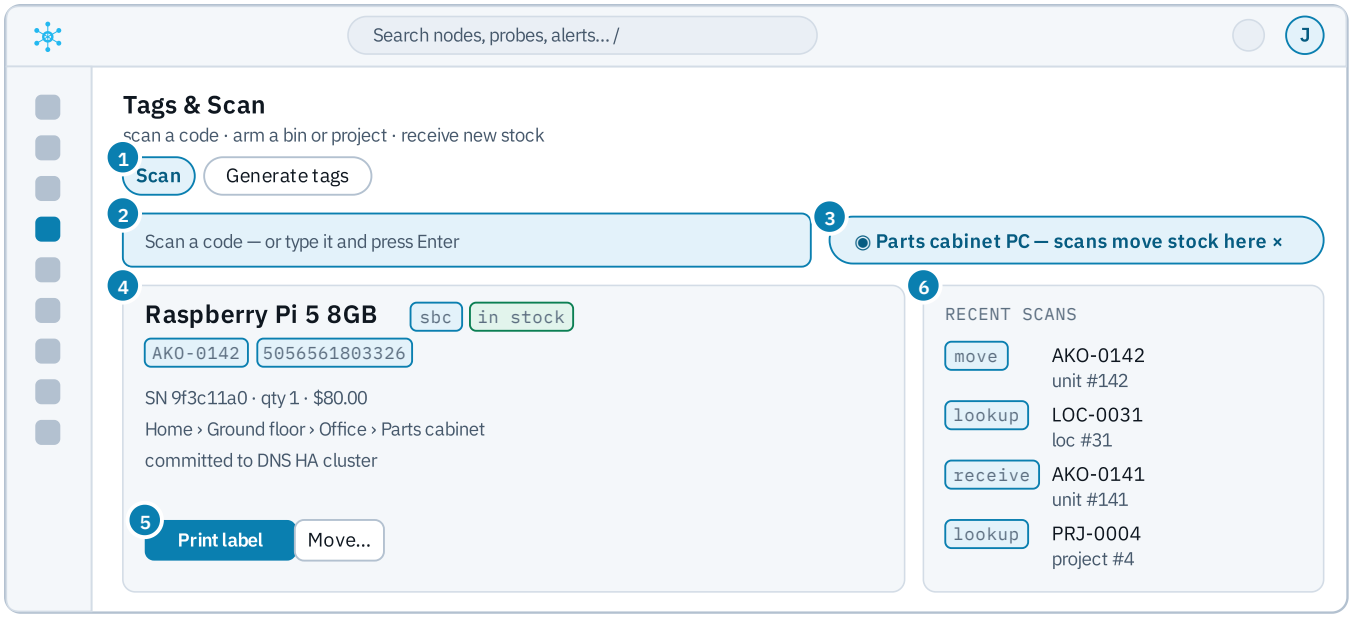


Figure 4.8 Tags & Scan with a bin armed. Every unit scanned now moves into the Parts cabinet.

#	CONTROL	WHAT IT DOES
1	‹**Scan**› · ‹**Generate tags**›	Scanning, or minting location tags (“Tag storage locations”).
2	Scan field	Always ready for a scan. It glows when something is armed.
3	Armed pill	Shows what is armed and what the next scan will do. ‹**×**› or `Esc` disarms.
4	Card	The thing you just scanned: an item, location, project or port, with its details.
5	‹**Print label**› · ‹**Move...**›	Card actions. Locations offer ‹**Arm as target**›; projects offer ‹**Arm project**› and ‹**Print tray label**›.
6	Recent scans	The last 25 scans with their action (lookup, receive, move, associate, print...) and age.

Ways to scan

- **Barcode scanner (recommended).** Any USB or Bluetooth scanner that acts as a keyboard, ending each code with Enter. Pair it with the iPad or computer like a keyboard. Scans are caught even when the cursor is not in the scan field, unless a dialog or the command palette is open.
- **Typing.** Type the code into the scan field and press `Enter`.
- **Camera.** The Tags & Scan page does not use the camera. RackWire's own scan dialog can, but only in browsers with a built-in barcode reader; Safari on iPad and iPhone has none. On Apple devices, use a scanner or type the code.

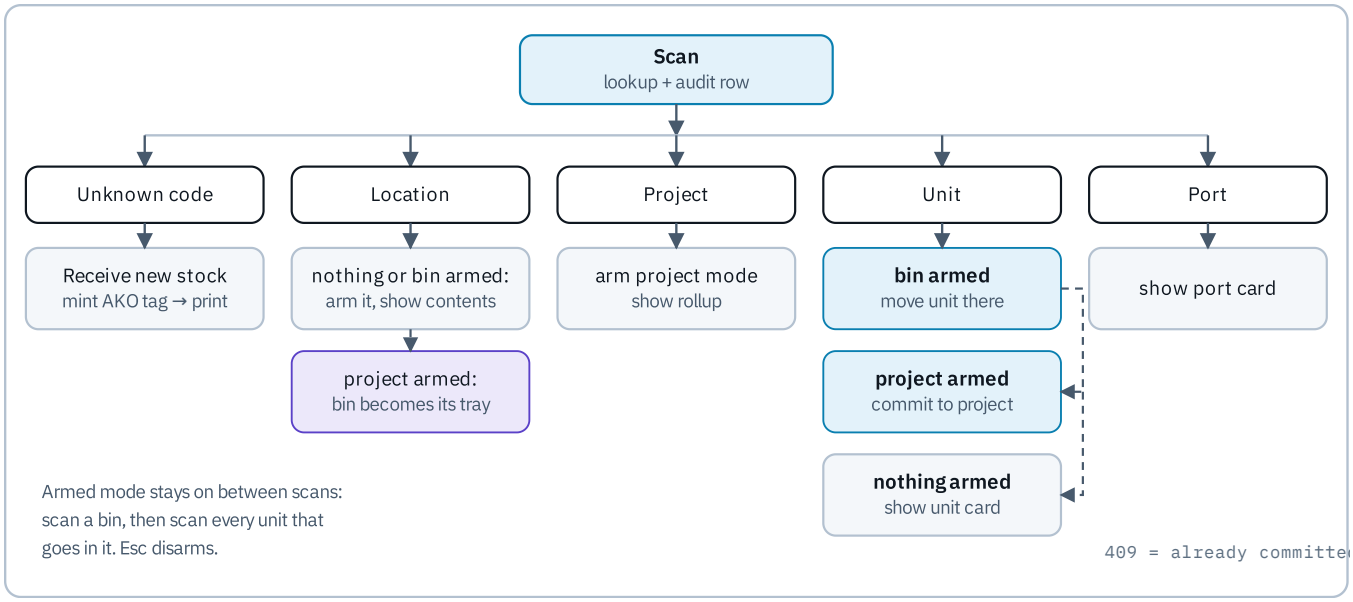


Figure 4.9 What a scan does depends on what you scanned and what is armed.

Put items away in a bin

1. Scan the bin's `LOC-` tag. Its card opens and the bin is armed: “scans move stock here”.

2. Scan each item's `AKO-` tag. Each one moves into the bin, and a message confirms it. The bin stays armed.
3. Press `Esc` when done.

Gather parts for a project

1. Scan the project's `PRJ-` tag. Project mode is armed: “scans commit to this project”.
2. If the project has no tray yet, scan a bin's tag; it becomes the project's tray.
3. Scan each item. It is committed to the project and moved to the tray.
4. Press `Esc` when done.

The dashboard cannot yet issue a `PRJ-` tag for a project. If a project has no tag, commit its items from the Projects tab instead (§4.6). ‹**Print tray label**› prints the tag of the project's tray bin; the bin needs a `LOC-` tag from ‹**Generate tags**› first.

Receive new stock

1. Scan the code on the new item (a serial number or the retail barcode). Because SolariNet does not know it, **Receive new stock** opens.
2. Choose the ‹**Model**› (or add one with ‹**+**›), check the ‹**Code type**› and ‹**Serial**›, and choose the ‹**Location**›. If a bin is armed it is preselected.
3. Enter ‹**Quantity**›, ‹**Unit cost (USD)**› and the ‹**Receipt**› if known.
4. Select ‹**Receive + tag**›. The item is created, the scanned code is stored with it, a new `AKO-` tag is issued, and the Print label dialog opens.

Tag storage locations

On ‹**Generate tags**›, every storage unit is listed with its path and tag. Select ‹**Tag**› on a row, or ‹**Tag all untagged**› to issue `LOC-` tags for every storage unit without one. Then select ‹**Print**›. Only storage units can be tagged here; add storage units on the Locations tab first.

> **IPAD AND IPHONE**
> Pair the scanner with the iPad in Settings → Bluetooth; it then works like a keyboard. Scans are caught even when the on-screen keyboard is hidden. On an iPhone, the recent-scans column leaves little room for the card; turn the phone sideways `INFERRED`.

4.8 Labels

Labels are printed from the **Print label** dialog, opened from an item, location or project card, or automatically after receiving stock.

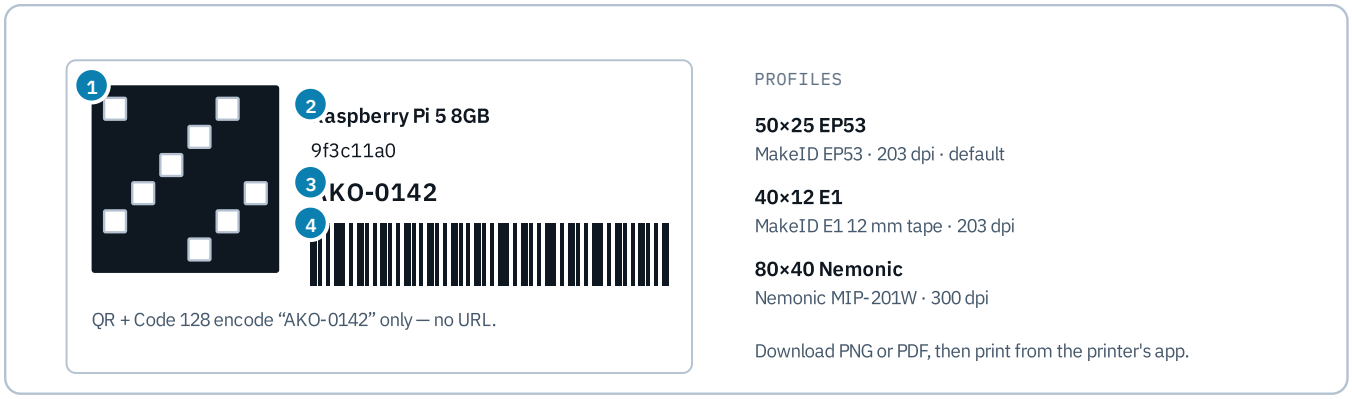


Figure 4.10 A 50 × 25 mm label and the three label profiles. Callouts: 1 QR code, 2 name line, 3 serial or detail line, 4 the tag code with Code 128 barcode.

1. Choose the ‹**Label profile**› to match your printer and tape.
2. Choose ‹**PNG**› or ‹**PDF**› and select ‹**Download**›.
3. Print the file from the printer's own app.
4. Select ‹**Mark printed**› so SolariNet records that the label exists. The dialog then shows when it was last printed.

The codes hold only the tag text, for example `AKO-0142`, not a web address. Any scanner, including a phone camera app, reads the code, but only SolariNet knows what it means. SolariNet cannot send labels straight to a printer.

> **!** **IF THE PREVIEW IS BLANK**
> If the dialog says “Label renderer unavailable”, the label tools are missing on the server. An administrator installs `python3-qrcode` and `python3-barcode` on xenon.

4.9 RackWire

PURPOSE	Plan network, power and interconnect: devices, ports, cables, circuits and PoE budgets, with live telemetry.
OPEN	Sidebar → ‹**RackWire**›
WHO	`VIEWER` to view; inventory lookups inside RackWire need `OPERATOR`

RackWire is a separate planner shown inside the dashboard. It follows the dashboard's light or dark theme. Its own manual covers drawing and planning. The dashboard adds:

CONTROL	DOES
‹**Network discovery**› · ‹**Inventory**›	Jump to those dashboard pages.
‹**Scan inventory barcode**›	Opens RackWire's scan dialog to place an owned item by its tag.
‹**Map**› · ‹**Table**›	Switch the planner view.
‹**Open standalone**›	Opens RackWire full-screen in a new tab.
Selected chips	When you select devices in the planner, chips show their names. Select a chip to centre the map on it; ‹**Clear**› deselects.

Place an owned item from its tag

1. Select ‹**Scan inventory barcode**›.
2. Scan the item's tag with a scanner, or type it, and select ‹**Look up**›.
3. RackWire looks for a match in this order: codes you have bound before, then the SolariNet inventory, then its own device library. A match is placed with its asset tag and serial.
4. If nothing matches, bind the code to a known model, or create a blank device with the code attached, and select ‹**Bind and place**›.

Items only appear with real port layouts if their hardware model has a RackWire profile (“RackWire device profiles”).

> **IPAD AND IPHONE**
> The planner needs room. On an iPhone, select ‹**Open standalone**› and turn the phone sideways. RackWire's camera scan does not work in Safari; use a scanner or type the tag.

CHAPTER 5

Building and changing the fleet

The Build group of the sidebar is where systems join SolariNet and where their configuration is changed: finding new machines, enrolling and installing them, applying software, planning maintenance, and setting fleet configuration. Most actions here change real machines. Read the warnings in each section before you use them.

> **!** ACTIONS ON THIS CHAPTER'S PAGES ARE REAL
> Reimage wipes a disk. Decommission erases data on a host. Retire stops monitoring. Scans send traffic across a network. Each is marked below. All of these pages are **Lab** maturity (§2.7): use them on the home lab only.

5.1 Network discovery

PURPOSE	Find machines on the network that are not yet monitored, and adopt them.
OPEN	Sidebar → ‹**Discovery**›
WHO	VIEWER to review candidates; OPERATOR to scan, ignore and adopt
MATURITY	Lab

Machines are found passively (mDNS, neighbour tables, LLDP, SNMP) and by active scans. Each one found becomes a **candidate** until you adopt it or ignore it. The heading counts candidates and every asset seen.

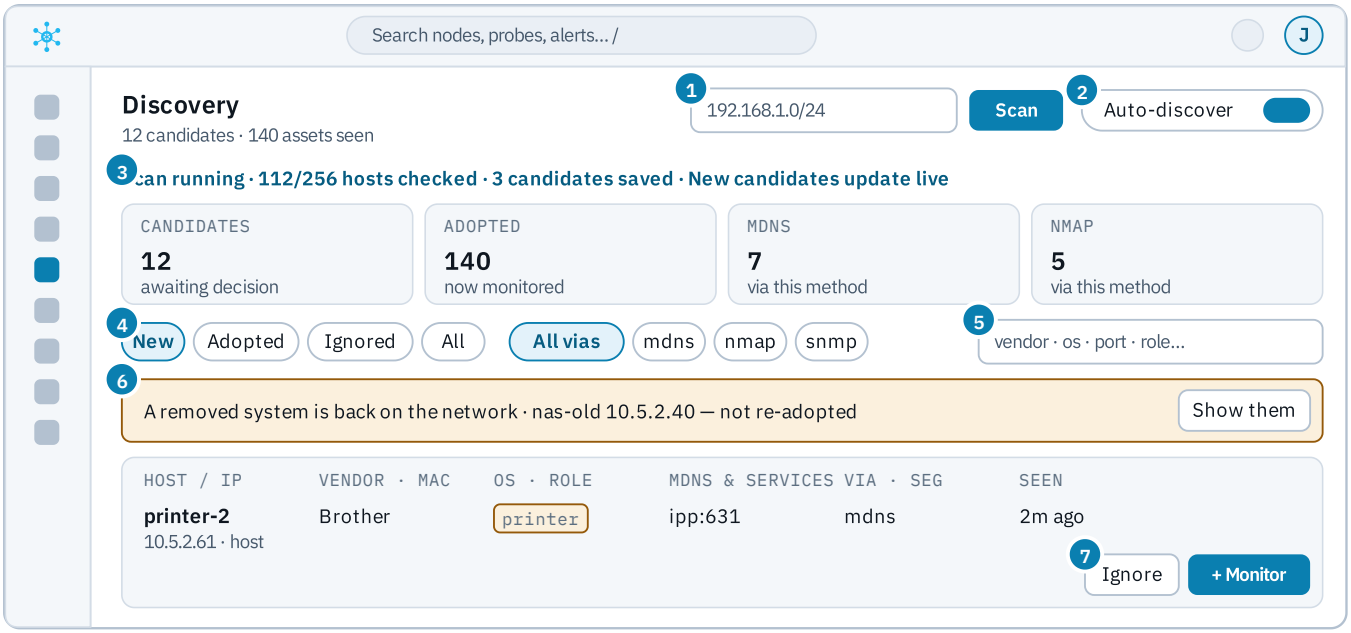


Figure 5.1 Network discovery with a scan in progress.

#	CONTROL	WHAT IT DOES
1	CIDR · ‹**Scan**›	Type a network range, for example `10.0.0.0/24`, and select ‹**Scan**› to probe it. `OPERATOR` only.
2	Auto-discover	Pauses or resumes background discovery. The switch is shown to everyone, but only an operator's change is accepted.
3	Progress line	While a scan runs: hosts checked, candidates saved. New candidates appear as they are found. The scan keeps running if you reload or leave the page.
4	Filter chips	‹**New**› (the default), ‹**Adopted**›, ‹**Ignored**›, ‹**All**›, and one chip per discovery method.
5	Search	Matches vendor, OS, port, role or segment.
6	Returned-system notice	Appears when a system you removed is back on the network (below).
7	‹**Ignore**› · ‹**+ Monitor**›	‹**Ignore**› hides a candidate under the Ignored chip. ‹**+ Monitor**› opens the adopt dialog.

Table columns: Host / IP, Vendor · MAC, OS · Role, mDNS & services, SNMP sysDescr, Neighbor (an LLDP chip names the switch port it is plugged into), Via · Seg (method and network segment) and Seen.

Scan a network range

1. Type the range in CIDR form. Anything else is refused with “Enter a CIDR, e.g. 192.168.1.0/24”.
2. Select ‹**Scan**›. The page confirms “Scan accepted — tracking progress in Discovery”.

3. Wait for “Scan complete — N host(s) found”.

Only private ranges can be scanned (10.x, 172.16–31.x, 192.168.x, loopback and link-local), and a range may hold at most 4,096 hosts, a /20. A larger range is refused, not trimmed; split it into several scans.

Adopt a system

1. Select ‹**+ Monitor**› on the candidate.
2. In **Monitor system**, check the ‹**Display name**› and choose a ‹**Classification**› (server, appliance, network, iot, host or other) and a ‹**Pool**›.
3. Under ‹**Services to monitor**›, untick any discovered service you do not want probed. All are ticked by default. If none were found, only the host heartbeat is used.
4. Leave ‹**Host heartbeat (ICMP ping)**› on unless the machine does not answer ping.
5. Select ‹**Monitor system**›. The page confirms “Adopted *name* — N target(s)”, and the system appears on Systems and Reachability.

Adopting starts monitoring from outside, by probes. To collect full metrics, also install the client (“Install the client on an existing machine”).

A removed system is back

If a system you retired or removed reappears at the same address, it is *not* re-adopted automatically. The notice says so; select ‹**Show them**› to list them. Restore the original system to resume monitoring it (“Retiring a host”), or purge it first if it really is a different machine on the same address (“Purge a deleted system”).

> □ IPAD AND IPHONE
> The candidate table scrolls sideways. Adopting is easier on an iPad, where the adopt dialog shows every discovered service at once. INFERRED

5.2 Provisioning

PURPOSE	Enrol new hosts, install the client, stage OS installs and check configuration convergence.
OPEN	Sidebar → ‹**Provisioning**›
WHO	OPERATOR
MATURITY	Lab. Network-boot installs remain operator-only and lab-only.

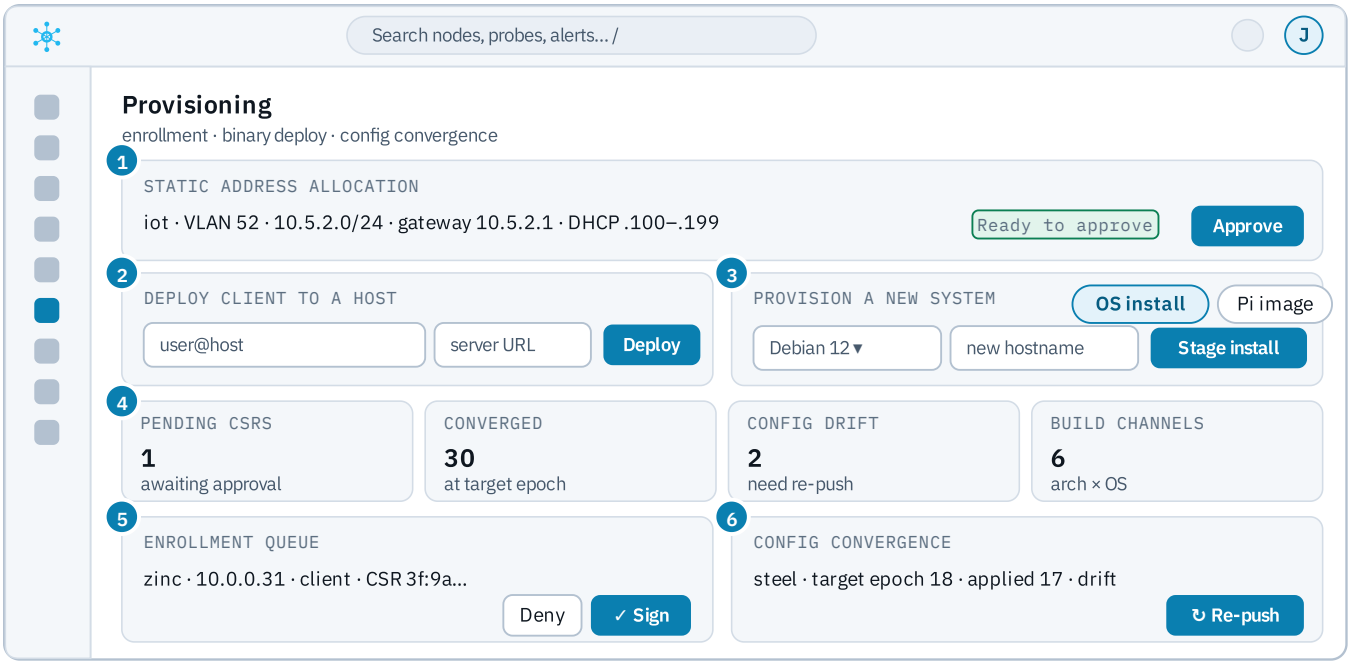


Figure 5.2 The Provisioning page.

#	PANEL	WHAT IT DOES
1	Static address allocation	Approves networks from which fixed addresses may be handed out (“Approve a network for fixed addresses”).
2	Deploy client to a host	Installs and enrols the SolariNet client on an existing machine over SSH.
3	Provision a new system	Stages a network OS install, or builds a Raspberry Pi image.
4	Summary tiles	**Pending CSRs** (hosts waiting to be signed), **Converged**, **Config drift** and **Build channels** (architecture × OS combinations with a client build).
5	Enrollment queue	Hosts asking to join. ‹✓ **Sign**› admits one; ‹**Deny**› refuses it.
6	Config convergence	Hosts whose configuration is behind the server. ‹↻ **Re-push**› sends it again. When none: “All nodes converged.”

Install the client on an existing machine

Before you start, the target must accept your SSH key and allow `sudo` without a password. For ARM hosts, a client binary for that architecture must already have been built.

1. In **Deploy client to a host**, type the login as `user@host`, for example `pi@10.0.0.254`.
2. Leave ‹**server URL**› empty unless told otherwise.
3. Select ‹**Deploy**› and watch the log under the form.

4. When it finishes, the host's certificate request appears in the Enrollment queue, or is signed automatically. It then starts reporting on Fleet Overview.

Admit a host

In the Enrollment queue, check that the host name, role and address are what you expect, then select ‹**✓ Sign**›. There is no confirmation step. The page confirms “CSR signed — *host* enrolled as *role*”. Select ‹**Deny**› for anything you do not recognise.

> **!** NOT WORKING YET
> ‹**+ Issue token**› shows a one-time code but does not register it with the server, so the code cannot be used. The ‹**Push**› button under binary builds only shows a message; it does not deploy anything. Use **Deploy client to a host** instead.

Install a new machine from the network

1. In **Provision a new system**, choose ‹**OS install**› or ‹**Pi image**›.
2. Choose the distribution (Debian 12, Ubuntu 24.04, openSUSE Leap 15 or Raspberry Pi OS), architecture, role (Sensor / client, Edge node or Appliance) and profile (Standard stack or Minimal). After switching mode, check the distribution again; it is not reset for you. INFERRED
3. Type the machine's network-card MAC address (or a host name) and the new host name.
4. Select ‹**Stage install**› (or ‹**Build image**›). The install is staged on benzene; the machine installs itself the next time it boots from the network.

For machines that are already in inventory, **Images & Stacks** is the better route: it checks architecture and image compatibility first (§5.3).

Approve a network for fixed addresses

The **Static address allocation** list shows networks known to the System of Record. A network can be approved only if it maps to exactly one live IPv4 subnet. Each row shows a state: *Ready to approve*, *Approved*, *Operator approval required*, *Approval needs review*, *Disabled at source*, or *Blocked* (missing subnet data, or overlapping prefixes).

1. Find the network with the search box.
2. Select ‹**Approve**› and confirm.

Any change to the network at its source revokes the approval, and it must be approved again. Approving here does not change the UniFi configuration. ‹**Revoke**› withdraws approval.

5.3 Images & Stacks

PURPOSE	Reinstall a machine from an OS image, or apply software stacks to it, in one deploy.
OPEN	Sidebar → ‹**Images & Stacks**›
WHO	VIEWER to look; OPERATOR for every action. Live data only: with no live connection the page shows nothing and every action is paused.
MATURITY	Lab

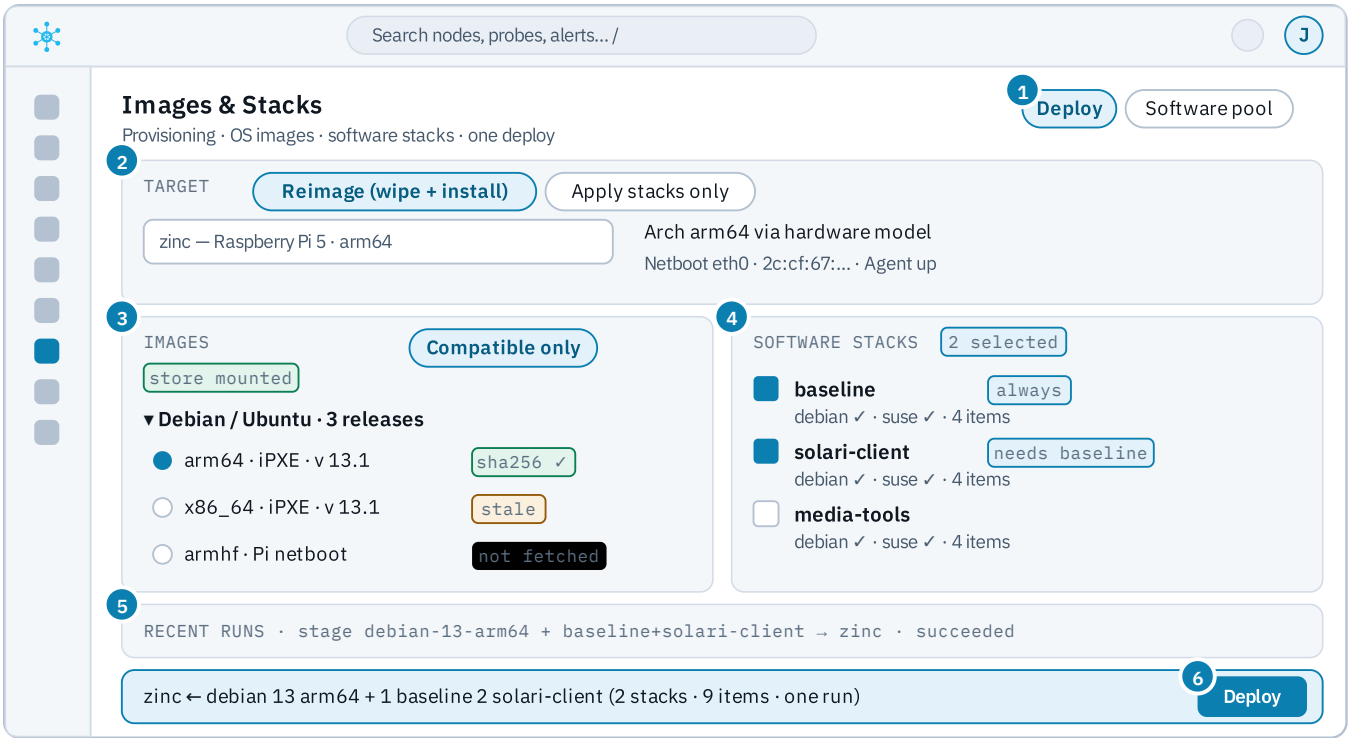


Figure 5.3 Images & Stacks, Deploy view. Choose a target, an image and stacks; the bar at the bottom says what will happen or what is blocking it.

#	AREA	WHAT IT DOES
1	‹**Deploy**› · ‹**Software pool**›	Deploy to a machine, or edit the software that stacks are built from.
2	Target	Mode (‹**Reimage (wipe + install)**› or ‹**Apply stacks only**›) and the machine, chosen from inventory. It shows the machine's architecture and where it came from, its network-boot interface, model, current OS, lifecycle and agent.
3	Images	OS images, grouped by distribution and release. Each shows architecture, boot method, version and freshness. ‹**Compatible only**› hides images that cannot be used on this target.
4	Software stacks	Tick the stacks to apply. Required and dependent stacks are added for you and labelled.
5	Recent runs	Every deploy, with status and a log. ‹**Follow (3 s)**› keeps the log updating.
6	Deploy bar	Summarises the deploy and holds the button. If something is missing, it names the first blocker instead.

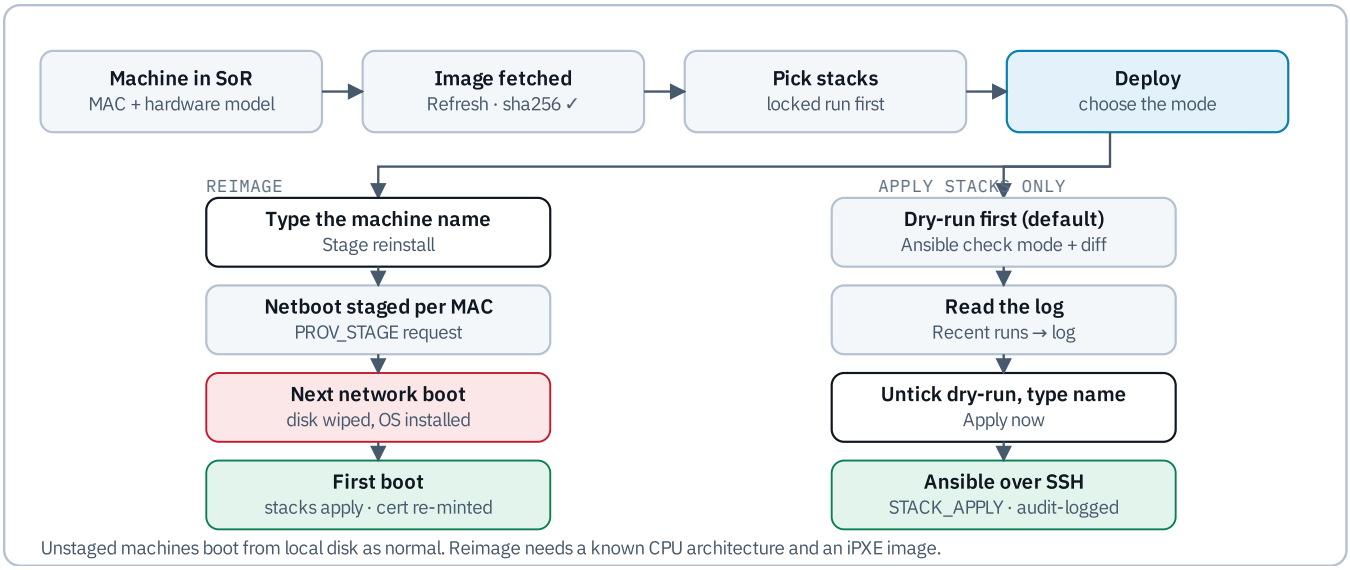


Figure 5.4 What a reimage or an apply does, from choosing the target to a host that reports in.

Choose the target

Only machines in inventory that have a MAC address are listed. If the architecture is unknown, select ‹**Set**› to give it; the setting applies to *every* machine of that hardware model. Architecture can also come from the machine's running agent.

Read the image list

SHOWN	MEANING
● sha256 ✓	Downloaded and checksum verified.
● stale	Due for refresh: a weekly image older than 8 days, or a point release older than 45 days.
● not fetched	Not downloaded. Select ‹**Refresh**› first.
netboot unverified	Network boot of this image has not been confirmed.
Greyed row	Cannot be used: wrong architecture, not fetched, or not an iPXE image (only iPXE images can be staged for network boot). The reason is shown.

‹**Refresh**› queues a download and confirms “Refresh of *image* queued … follow it in Recent runs”. It is refused while the image store is not mounted; the store's status and free space show at the top of the list. ‹**Add image**›, ‹**Edit**› and ‹**Delete**› manage entries; deleting asks you to type `delete`.

Choose software stacks

A stack is a named set of software, such as a monitoring client or a DNS server. Tags on each stack say *always* (applied every time), *needs X*, or *auto · required by selection*. A green family tag means the stack has packages for that OS family; a red one means it does not, and a deploy to that family is refused. Stacks run in a fixed order: locked stacks first, then dependencies, then their set order.

Reimage a machine

> ! **DESTROYS DATA**
> Reimage wipes the machine's install disk the next time it boots from the network. Back up anything you need first.

1. Set the mode to ‹**Reimage (wipe + install)**› and choose the machine.
2. Choose a usable image and the stacks to install afterwards.
3. Check that the deploy bar shows no blocker, then select its button.
4. In **Stage reinstall of *host*?**, read the summary, type the host name, and select ‹**Stage reinstall**›.
5. The page confirms “Reinstall of *host* staged on *MAC*”. Reboot the machine into network boot. It installs, gets a new node certificate at first boot, and reports in.

Apply stacks to a running machine

1. Set the mode to ‹**Apply stacks only**›, choose the machine and tick the stacks.
2. Leave ‹**Dry-run first**› ticked and select ‹**Run dry-run**›. This runs Ansible in check mode and shows what would change, without changing anything.
3. Read the dry-run log in Recent runs.

4 Untick ‹**Dry-run first**›, select the button, type the host name, and select ‹**Apply now**›.

Deploy bar blockers, in the order they are reported

MESSAGE (SHORTENED)	FIX
Live API unavailable	Wait for the dashboard to reconnect.
Operator role required	Sign in with an operator account.
Choose a target	Pick a machine.
Dependency cycle	Two stacks need each other. Fix them in the Software pool.
No stacks	Tick at least one stack.
Conflicting entries	Two selected stacks set the same thing differently. Untick one.
Architecture unknown	Select ‹**Set**› on the target.
No MAC	Add the network card's MAC to the machine in inventory.
Image not deployable · Select an image	Choose a usable image (reimage only).

Software pool

The Software pool lists every piece of software stacks can contain. Each entry has a logical name, a type (pkg, repo, artifact, build, script or role), the package name for each OS family (SUSE, Debian / Ubuntu, Fedora), an optional version pin, and the stacks that use it. Entries are stored in the System of Record and turned into Ansible settings at run time; no YAML is edited here. Environment variable names that look like secrets are refused.

> **IPAD AND IPHONE**
> Below 1,050 pixels wide, as on an iPad in portrait, the panels stack in a single column with the deploy bar still at the bottom. On a phone, the log and edit drawers fill the screen. Reimaging from an iPhone is possible but not recommended: the target and image details are hard to check on a small screen.

5.4 Maintenance windows

PURPOSE	Silence alerting while you work on a host or on the whole fleet.
OPEN	Sidebar → ‹**Maintenance**›
WHO	OPERATOR to schedule and cancel
MATURITY	Lab

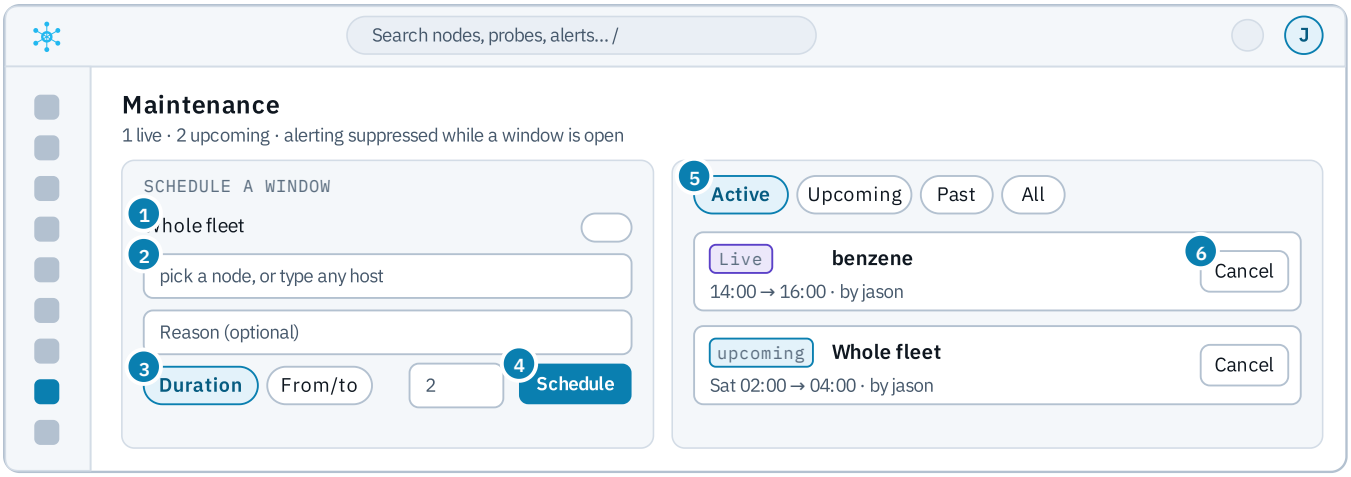


Figure 5.5 Scheduling a maintenance window.

#	CONTROL	WHAT IT DOES
1	Scope	On: the whole fleet. Off: one host.
2	Host	Pick a node, or type any host name, for example *benzene*. Add an optional reason.
3	‹**Duration**› · ‹**From / to**›	Hours from now (in steps of half an hour; default 2), or a start and end time.
4	‹**Schedule maintenance**›	Creates the window.
5	Filter chips	‹**Active**›, ‹**Upcoming**›, ‹**Past**›, ‹**All**›.
6	‹**Cancel**›	Ends or removes a window after “Cancel?” → ‹**Yes**›.

1. Choose the scope and, for one host, the host.
2. Choose ‹**Duration**› and enter the hours. Prefer Duration to From / to (see the note below).
3. Select ‹**Schedule maintenance**›. The window appears under Active or Upcoming, with who created it.
4. When the work is done, select ‹**Cancel**› → ‹**Yes**› to resume alerting early.

> **TIMES**
> The list shows times in your device's time zone, but ‹**From / to**› times are read in the server's time zone. On a device set to another zone, the window lands at the wrong time. The form also does not check that the end is after the start. Use ‹**Duration**› and check the times shown afterwards.

While a window is open, alerts for its hosts are suppressed. Other pages do not yet show a maintenance badge, so check this page to see what is silenced.

5.5 Config & Rules

PURPOSE	Set fleet-wide and per-host configuration, then push it to the fleet.
OPEN	Sidebar → ‹**Config & Rules**›
WHO	OPERATOR to save and push. The controls are not disabled for viewers; a viewer's save is refused.
MATURITY	Lab

The page has two tabs, ‹**Global**› and ‹**Per-agent**›, plus ‹**Alert rules**› (“Create a rule”). Changes on the Global tab take effect when you select ‹**✓ Save & push**›, which sends them to every host. The page confirms “Config saved → pushed to fleet (solariCtl)”.

Global settings

GROUP	SETTING	RANGE
Sampling	Sample interval: how often clients report	5–120 s
	Watchdog	1–30 s
Retention	History kept	7–365 days
	Monthly partition pruning	on / off
Probe defaults	Round interval	10–120 s
	Probes per round	1–20
	Replication factor: how many monitors probe each target	1–5
	Peer gossip	5–60 s
Failover lease	Renew	1–30 s
	Lease TTL	5–60 s
Discovery & enrolment	Auto-discovery; auto-enrol discovered hosts (only when approval is not required)	on / off
Read only	Internal CA, Transport ports, Identity & SSO. These come from the server's configuration file and cannot be changed here. ‹**Test sign-in**› checks the SSO provider.	—

Per-host settings

The ‹**Per-agent**› tab lists hosts by role: ‹**Server**›, ‹**Network**› (monitors) and ‹**Application**› (clients, the default). ‹**Drift only**› shows hosts whose configuration is behind; ‹**Retired**› shows retired hosts. Select a host to open its settings.

ROLE	SETTINGS
Client	Sample interval; watched applications; log watch pattern (default `(error|fail)`).
Monitor	Round interval; probes per round; replication factor; the probe targets assigned to it.
Server	Lease renew; lease TTL; database pool size; bind addresses.

Select ‹✓ **Push to node**› to send the host its settings.

Retire or decommission a host

ACTION	EFFECT	CONFIRM
‹**Retire**›	Stops monitoring and removes the host from the active roster. Nothing on the host is changed. It can be restored.	Select ‹**Retire node**›.
‹**Decommission**›	Securely wipes the selected data from the host over the control plane, then retires it. **Cannot be undone.**	Choose what to wipe (configuration, certificates, spool, logs, service unit), then type the host name.

After a decommission request, the page says “Decommission request accepted for *host*; waiting for node acknowledgement”. The host must be online to receive it.

CHAPTER 6

Infrastructure views

The Infra group of the sidebar shows the services the rest of SolariNet depends on: the Git server, the certificate authority and DNS. These pages are read-only. This chapter also explains how a change made in the dashboard reaches DNS.

6.1 Git

PURPOSE	See the state of the Forgejo Git server on cesium at a glance.
OPEN	Sidebar → ‹**Git**›
WHO	VIEWER. Read-only.
MATURITY	Lab

Summary tiles count **Repositories**, **Open PRs** and **Open issues**, and show the **Server** version and state. Below them is one card per repository, then **Recent commits** and **Open pull requests**. Select a repository card to open it in Forgejo in a new tab; sign in there with Akoria SSO. If the server cannot be reached the page says “Forgejo unavailable”.

6.2 Certificates

PURPOSE	Check the Akoria certificate authority, the certificates it has issued to SolariNet hosts, and revocation status.
OPEN	Sidebar → ‹**Certificates**›
WHO	VIEWER. Read-only.
MATURITY	Lab

The heading names the CA, the step-ca service on chlorine. Summary tiles show whether **step-ca** is healthy or unreachable, and count **CA certificates**, **Provisioners** (the ways the CA will issue certificates) and **Node certs**.

- **CA certificates.** Each shows an expiry pill: ● **N d left**, turning ● **amber** under 60 days, ● **expired**, or ● **unknown**.
- **Fleet mTLS certificates.** One row per enrolled host: Host, Cert CN, Role, Arch, Enrolled and Last seen. These are the client certificates hosts use to talk to the server. A host gets one when it is signed in Provisioning (“Admit a host”) and a new one after a reimage.
- **CRL distribution.** The current certificate revocation list generation (“CRL #N”) and whether each host has picked it up: *Current*, *Stale receipt* (holds an older list), *No receipt*, *Offline* or *Expired generation*. A revoked certificate stops working only once a new list is generated and the hosts have it; until then the page notes that the revocation is not yet effective.

If an expiry turns amber, tell the administrator. Certificates are renewed outside the dashboard.

6.3 DNS

PURPOSE	Ask each DNS server the same question, check that the servers agree, and browse the records SolariNet generates.
OPEN	Sidebar → ‹**DNS**›
WHO	VIEWER. Read-only: records are changed by editing the host or network they belong to, not here.
MATURITY	Lab

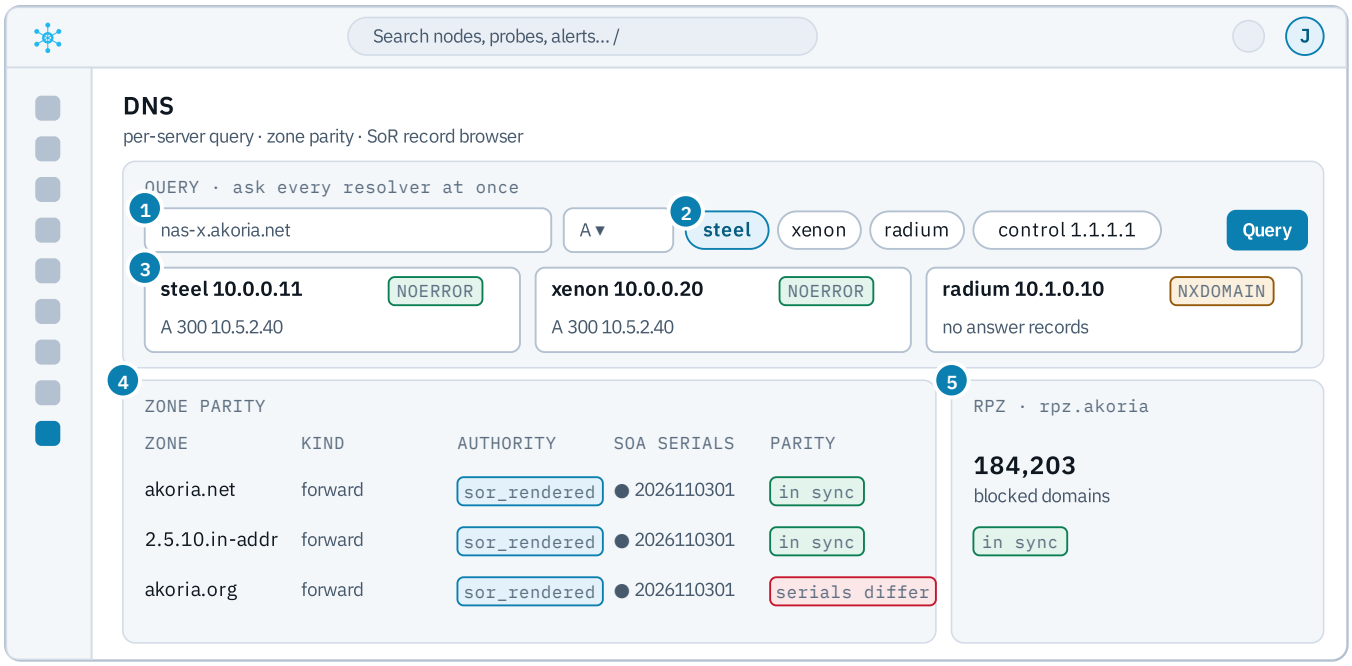


Figure 6.1 The DNS page: a query sent to every server, then zone parity.

#	CONTROL	WHAT IT DOES
1	Name · type	The name to look up and the record type (A, AAAA, CNAME, PTR, TXT, MX, SRV, NS, SOA or ANY). Canary chips fill in standard checks: *xenon.akoria.net A*, *radium.akoria.org A* and *doubleclick.net TXT* (a name the ad-blocking list should block). The last ten queries are kept.
2	Server chips	Which servers to ask: steel, xenon, radium, and 1.1.1.1 as a public control. The control is never counted when checking agreement.
3	Results	One card per server with its answer. ● **NOERROR** means it answered; ● **NXDOMAIN** means the name does not exist there. A card marked *differs* disagrees with the others; *RPZ BLOCKED* means the name is on the block list.
4	Zone parity	Each zone with its kind, the server that holds the master copy, every server's SOA serial, and *in sync* or *serials differ*. Updated every 60 seconds.
5	RPZ tile	The ad- and tracker-blocking zone *rpz.akoria* and how many domains it blocks.

Check that a name resolves everywhere

1. Type the name, for example *benzene.akoria.net*, and choose the type.
2. Keep all servers selected and run the query.
3. Compare the cards. If steel and xenon disagree, a bar says so and points to zone parity: a zone transfer has not happened yet, or has failed.

If the query fails with “Query failed — the DNS API is unreachable or errored”, your session may have expired. Reload the page and sign in again.

Browse SoR records

SoR records lists the records held in the System of Record, by zone. Filter by name or data, by type, and by origin: *generated* records are derived from host and address data, so to change one, edit the host. *Explicit* records were entered directly. Up to 500 rows are shown. **Render preview** shows the zone file exactly as the generator will produce it.

> □ IPAD AND IPHONE
> Result cards wrap onto more rows on a narrow screen `INFERRED`. Zone parity and SoR records tables scroll sideways.

6.4 How a change reaches the network

The System of Record (MariaDB on cesium) is the single place where hosts, addresses, networks, DNS records and inventory are kept. Changing it from the dashboard starts an automatic chain. Three services on xenon run the chain.

STEP	WHAT HAPPENS	SERVICE
1	You save a change, for example adopting a host or editing an address. The database records an event in the same transaction, so no change is lost.	SoR on cesium
2	The event is published on the message bus (RabbitMQ on benzene).	`sor-emit`
3	The DNS applier regenerates the zones from the current records, releases them on xenon, then refreshes each secondary server and checks that its serial matches. It also re-checks everything every hour.	`sor-apply-dns`
4	In the other direction, systems you adopt in Discovery are added to the SoR by address, and then flow forward to DNS like any other change.	`sor-reconcile`

A new host should resolve within a minute or two of being adopted `INFERRED`. Check with the DNS page; if it does not, see Chapter 7.

CHAPTER 7

Troubleshooting

Find the symptom, then follow the steps in order. If none of them help, report the problem (§8.2).

7.1 Signing in

SYMPTOM	CAUSE	WHAT TO DO
Sent back to the sign-in screen while working	Your session ended. Sessions end after 30 minutes without activity, and in any case after 1 hour (SSO) or 8 hours (local account).	Sign in again. Unsaved form contents are lost, so save often during long edits.
Local sign-in refused after several tries, even with the right password	Local sign-in allows 5 attempts per account and device address in any 15 minutes. Further attempts are refused.	Wait 15 minutes, then try once, carefully.
Akoria SSO shows an error before asking for your passkey or code	The SSO server could not check your password with the directory. This has happened when a directory account's password expired.	Try a local account if you have one, and tell the administrator. It is not something you can fix from the dashboard.
Signed in, but changes fail with “This action requires an authenticated operator role.”	Your account is a viewer.	Ask the administrator to add you to the operator group. Your role is rechecked about every minute, so the change takes effect without signing out.
The profile chip says “Local account” although you used SSO	A display fault.	Ignore it. Your role is still taken from SSO.

7.2 Demo data is showing

The red “DEMO DATA — live API unreachable” banner means the dashboard could not load live data when it started and is showing a built-in sample fleet. Nothing you see is real, and changes do not work. The reason is in brackets.

REASON IN BRACKETS	WHAT TO DO
authentication required	You are not signed in. The sign-in screen should be showing; sign in.
forced offline	The page was deliberately opened in offline mode. Open `https://dashboard.akoria.net/` again from its normal address or Home Screen icon.
A network or HTTP error	Check that the device is on the home network or VPN, then reload. If other devices also see the banner, the dashboard server (xenon) or its database is down; tell the administrator.

The banner never clears by itself. After fixing the cause, reload the page.

7.3 Data looks wrong or old

- **DATA STALE banner.** Live updates have stopped. Wait a minute; if it stays, reload. Changes are blocked until the data is live again.
- **PARTIAL DATA banner that will not go away.** It does not clear itself when the missing sections return. Reload.
- **A system shows Unknown or is dimmed.** It has stopped reporting. Check it on Reachability; if it is retired, it only appears under the ‹**Retired**› chip.
- **The topology map looks unfamiliar.** If topology data fails to load, the map falls back to the sample layout without saying so. Compare it with Fleet Overview.
- **An action says “queued”.** The server daemon did not reply within a few seconds. The action will still run; check again in a minute.
- `control_unavailable`**.** The server daemon cannot be reached. Nothing was queued. Tell the administrator.

7.4 Inventory and scanning

SYMPTOM	WHAT TO DO
Scans do nothing	Close any open dialog or the command palette: scans are not caught while one is open. Check that the scanner is paired as a keyboard and ends each code with Enter. Try typing the code into the scan field.
Every scan fails	Scanning needs OPERATOR and a live connection.
The camera does not work on iPad or iPhone	Expected. Safari has no built-in barcode reader. Use a scanner or type the code (“Ways to scan”).
A retail barcode is refused when attached	The check digit is wrong, usually a mis-scan. Scan it again, or choose ‹**SKU**› or ‹**Other**› if it is not a GTIN.
“… is already attached to …”	A code belongs to one product type. Remove it from the other type first if it belongs here.
A receipt disappeared after saving	Receipts saved without a scan file are not stored. Upload it again with a scan or photo.
“This was changed elsewhere since you opened it”	Someone else saved first. The latest version has been loaded; check it and save again.
A save fails with 409 and “try again”	The database was briefly busy with a conflicting change. Nothing was saved; repeat the action.
“Label renderer unavailable”	The label tools are missing on the server; tell the administrator (§4.8).
A project has no tag to scan	Project tags cannot be issued from the dashboard yet. Commit items from the Projects tab.

7.5 A host does not resolve

1. On the DNS page, query the name against all servers (“Check that a name resolves everywhere”).
2. If every server says NXDOMAIN, check that the host exists in the SoR with an address: look under **SoR records** for a generated record.
3. If xenon answers but a secondary does not, check **Zone parity**. Serials that differ mean the secondary has not taken the latest zone yet. Wait a few minutes and query again.
4. If they still differ, tell the administrator. The secondary refresh service on xenon may have failed.

7.6 Alerts and notifications

- **No phone message for an alert.** Only critical and warning alerts are sent to phones and email; info alerts are not. Web push reaches no browsers yet. Check whether a maintenance window covers the host.
- **Too many messages from one host.** There is no grouping or quiet hours. Schedule maintenance for the host, or lower its criticality, while you fix it.

- **The panel is red but the Alerts page looks calm.** The panel also alarms on a pool score of 100 or more and on any tier-4 (vital) system being down. Check Fleet Overview (“The alarm”).
- ‹**Ack all**› **counts more alerts than the tiles.** It also counts alerts that cleared in the last hour.

7.7 The app itself

- **The first load after an update is slow.** The dashboard prepares its code in the browser and keeps the result. Later loads are fast.
- **A change was deployed but you see the old version.** Reload. The app always fetches fresh code when the server is reachable, and only uses its saved copy when it is not.
- `/` **opens the command palette while typing.** This happens in some multi-line and selection fields. Press `Esc` and carry on.
- **Something is cut off on iPhone.** Turn the phone sideways, or use an iPad. Wide tables scroll sideways.

7.8 Known issues

These faults are known and recorded. Each has a workaround above or in the section named.

ISSUE	WORKAROUND	SECTION
Receipt without a scan is not stored	Always attach a scan	§4.5
‹**+ Issue token**› codes cannot be used	Deploy the client over SSH	“Install the client on an existing machine”
Binary ‹**Push**› only shows a message	Deploy the client over SSH	“Install the client on an existing machine”
Survey now and Push config on Node detail act fleet-wide	None needed; harmless	§3.2
Rule threshold slider ignores keyboard changes	Drag the slider	“Create a rule”
PARTIAL DATA banner does not clear	Reload	§2.6
Profile chip says Local account for SSO users	Ignore	§7.1
Web push reaches no browsers	Rely on iMessage and email	§3.9
Ack all count includes recently cleared alerts	Ignore	“Acknowledge alerts”
Topology falls back to sample data without warning	Compare with Fleet Overview	§3.6
Maintenance From / to uses the server's time zone	Use Duration	§5.4
Other pages show no maintenance badge	Check the Maintenance page	§5.4
Projects cannot be edited after creation	Set status and tray when creating	§4.6
Project tags cannot be issued	Commit from the Projects tab	“Gather parts for a project”
`/` opens the palette inside some fields	Press Esc	Appendix C
Some discovery details from nmap scans are not shown	None	§5.1

7.9 Questions

Is anything I do here sent outside the house?

No. Barcode lookups check only Akoria's own catalog. Notifications go to the household's own phones and email.

Can I break something by looking?

No. Reading never changes anything. Changes need the operator role, and the dangerous ones (reimage, decommission, merge, purge) ask you to type a name to confirm.

What is the difference between retire, decommission and purge?

Retire stops monitoring and can be undone. **Decommission** also wipes SolariNet's data from the host and cannot be undone. **Purge** (admin only) permanently deletes the record of a system that has already been deleted.

Why does a viewer see buttons they cannot use?

Some screens do not hide controls by role. The server still refuses the action.

Where are licence keys kept?

Not in SolariNet. Inventory records only a reference to where a licence is kept.

Does the dashboard work offline?

It still opens, but shows only the sample fleet, and every change is blocked.

CHAPTER 8

Getting help

The dashboard has no built-in help page. Help comes from this manual, its two companions, and the administrator.

8.1 The SolariNet manuals

MANUAL	FOR	COVERS
User's Manual (this one)	Everyone with dashboard access	Every screen and how to use it, troubleshooting and the glossary.
Developer's Manual	Anyone changing SolariNet	Building the C programs, the code layout, the daemons, the database schemas and deployment.
API Expansion Manual	Anyone scripting against SolariNet or adding to it	The HTTP API, its roles and errors, and how to add endpoints.

All three are published together on the documentation portal on xenon, as web pages and as PDF files for printing. Their source files are kept in the SolariNet repository under `docs/manuals/`. Open the repository from the Git page (§6.1).

> **USING THIS MANUAL**
>
> - On a wide screen the contents list is always shown at the left. On a narrower screen, such as an iPad in portrait, the contents button at the top left opens it.
> - Type in the search box to highlight every match. Press `Enter` to jump to the next one. The contents list hides chapters with no matches.
> - The theme button switches between automatic, light and dark.
> - The print button prints on US Letter paper, one chapter per new page.

8.2 Reporting a problem

Before reporting, check Chapter 7 and the known issues (§7.8). Then send the administrator:

1. **What you were doing**: the page, and the button or field you used.
2. **What happened**: the exact message, copied or photographed, and any banner at the top of the page.
3. **When**: the time, to the minute. Server logs are found by time.
4. **Where**: the device and browser, for example iPad, Safari.
5. **Who**: the account you were signed in as, and its role if you know it.

Problems with SolariNet itself are tracked as issues in the SolariNet repository on the Akoria Git server. Never include passwords, codes or keys in a report or a screenshot.

> **URGENT**
> If a vital (tier-4) system is down, the LED panel alarms on its own (“The alarm”). Still tell the operator if you are not sure they have seen it. If the dashboard itself will not load from any device, contact the operator directly.

APPENDIX A

Glossary

Acknowledge (ack)	Mark an alert as seen. It stays open until its cause clears, but stops counting as new.
Adopt	Start monitoring a discovered system.
Agent · client	The SolariNet program installed on a host. It reports metrics to the server.
Akoria	The household's network and its domains, *akoria.net* and *akoria.org*.
Allocated · deployed	Stock statuses: reserved for a project, or installed and in use.
Armed	On Tags & Scan, a bin or project selected as the target for the next scans.
AXFR	A DNS zone transfer: how a secondary server copies a zone.
CA · step-ca	The certificate authority that issues Akoria's internal certificates. It runs on chlorine.
Catalog · product type	The list of kinds of thing you own, each with its barcodes.
CDC	Change data capture: the System of Record notes every change so it can be sent on to DNS and other systems.
CIDR	A way of writing a network range, such as `10.0.0.0/24`.
Converged · drift	A host is converged when its configuration matches the server's; drift means it is behind.
CRL	Certificate revocation list: the list of certificates that must no longer be trusted.
CSR	Certificate signing request: a new host asking to be admitted.
Criticality tier	How much a system matters, from 0 to 4. Tier 4 is vital; its outage sets off the LED panel.
Decommission	Wipe SolariNet's data from a host and retire it. Cannot be undone.
GTIN · UPC · EAN	The standard retail barcode numbers.
iPXE · netboot	Starting a machine from the network so it can be installed without a USB stick.
Lifecycle	A system's state: active, retired or deleted.
Maintenance window	A period during which alerting is silenced for a host or the fleet.
Monitor	A host that probes other systems from the network.
mTLS	Mutual TLS: both ends of a connection prove who they are with certificates.
NXDOMAIN	The DNS answer meaning "no such name".
OIDC · SSO	Single sign-on through Akoria's Keycloak server on radium.
Opie	The automated analysis that writes incident reports.
Passkey	A sign-in credential kept on your device, unlocked with Face ID, Touch ID or a PIN.
Pool	A named group of systems, such as *DNS*.
Probe	A check by a monitor that a service answers.
Purge	Permanently delete a deleted system's record. Admin only.

PWA	Progressive web app: the dashboard added to the Home Screen.
Reimage	Wipe a machine and reinstall its operating system.
Retire	Stop monitoring a system without deleting it. Can be restored.
RPZ	Response policy zone: the DNS block list for ads and trackers.
SOA serial	A zone's version number. Servers that agree have the same serial.
SoR · System of Record	The MariaDB database on cesium that holds hosts, addresses, DNS and inventory.
Stack	A named set of software applied to a host.
Storage unit	A location that holds things: a drawer, bin, shelf or cabinet.
Tag	An Akoria label code: `AKO-`, `LOC-` or `PRJ-`.
Tombstone	The record a deleted system leaves behind, kept for history until purged.
Tray	The storage unit where a project's parts are gathered.
Wedge scanner	A barcode scanner that types what it reads, like a keyboard.

APPENDIX B

Permissions

What each role can do. The server enforces every row, whether or not the screen hides the control.

ACTION	VIEWER	OPERATOR	ADMIN	SECTION
Read any page	✓	✓	✓	—
Look up a barcode in the Catalog	✓	✓	✓	“Look up a barcode”
Acknowledge alerts; create and edit rules	—	✓	✓	§3.7
Control the LED panel	—	✓	✓	§3.10
Survey, push config, retire, restore	—	✓	✓	§3.2
Rename systems; create and delete pools; set criticality	—	✓	✓	§3.4
Purge a deleted system	—	—	✓	“Purge a deleted system”
Change inventory: components, catalog, locations, receipts, projects	—	✓	✓	Chapter 4
Scan on Tags & Scan; print and mark labels	—	✓	✓	§4.7
Look up inventory from RackWire	—	✓	✓	§4.9
Scan networks; pause discovery; adopt; ignore	—	✓	✓	§5.1
Deploy client; sign or deny; stage installs; approve networks	—	✓	✓	§5.2
Reimage; apply stacks; manage images and software	—	✓	✓	§5.3
Schedule and cancel maintenance	—	✓	✓	§5.4
Save and push configuration; decommission	—	✓	✓	§5.5
Change DNS records, certificates or Git	—	—	—	Not in the dashboard

Viewer experience is described from the server's rules. It has not been exercised with a viewer account in a browser.

APPENDIX C

Keyboard shortcuts

These work with a desktop keyboard and with an iPad keyboard.

KEYS	WHERE	DOES
⌘ K (Ctrl K on a PC) or /	Anywhere in the dashboard	Opens the command palette.
↑ ↓ Enter	Command palette	Choose and open a result.
Esc	Palette or dialog	Closes it.
Esc	Tags & Scan	Clears the scan field and disarms.
Enter	Tags & Scan scan field	Submits a typed code.
Enter	This manual's search box	Jumps to the next match.

Known issue: / also opens the palette while you type in some multi-line boxes and drop-down menus. Press Esc and carry on.

APPENDIX D

Coverage matrix

Every screen and function, and where this manual covers it. ✓ in the iPad column means the section has a device-specific note; otherwise the general guidance in §2.8 applies.

SCREEN OR FUNCTION	SECTION	FIGURE	PROCEDURES	IPAD
Sign-in, SSO, passkeys	§2.1	Figure 2.1	✓	—
Home Screen app	§2.2	—	✓	✓
App shell and navigation	§2.3	Figure 2.2 Figure 2.3	—	—
Command palette	“Command palette”	—	—	—
Roles and data banners	§2.5 §2.6	—	—	—
Fleet Overview	§3.1	Figure 3.1	—	✓
Node detail	§3.2	Figure 3.2	✓	—
Criticality	§3.3	—	—	—
Systems and system detail	§3.4	Figure 3.3	✓	—
Reachability	§3.5	Figure 3.4	—	✓
Topology Map	§3.6	Figure 3.5	—	—
Alerts and rules	§3.7	Figure 3.6 Figure 3.7	✓	—
Analysis (Opie)	§3.8	—	—	—
Notifications	§3.9	—	—	—
LED status panel and Panel page	§3.10	Figure 3.8 Figure 3.9	✓	—
Inventory concepts	§4.1	Figure 4.1	—	—
Components, entity links, RackWire profiles	§4.2	Figure 4.2	✓	✓
Catalog and barcode lookup	§4.3	Figure 4.3 Figure 4.4	✓	—
Locations, move, merge, container types	§4.4	Figure 4.5 Figure 4.6	✓	✓
Receipts	§4.5	—	✓	—
Projects	§4.6	Figure 4.7	—	—
Tags & Scan, scanning methods	§4.7	Figure 4.8 Figure 4.9	✓	✓
Labels	§4.8	Figure 4.10	✓	—
RackWire	§4.9	—	✓	✓

SCREEN OR FUNCTION	SECTION	FIGURE	PROCEDURES	IPAD
Discovery	§5.1	Figure 5.1	✓	✓
Provisioning	§5.2	Figure 5.2	✓	—
Images & Stacks, Software pool	§5.3	Figure 5.3 Figure 5.4	✓	✓
Maintenance	§5.4	Figure 5.5	✓	—
Config & Rules, per-agent, decommission	§5.5	—	—	—
Git	§6.1	—	—	—
Certificates	§6.2	—	—	—
DNS	§6.3	Figure 6.1	✓	✓
SoR sync to DNS	§6.4	—	—	—
Troubleshooting and known issues	Chapter 7	—	✓	—
Help and reporting	Chapter 8	—	✓	—

APPENDIX E

About this document

E.1 Sources and method

This manual was written from the dashboard and server source code on the main branch as of 3 November 2026, and checked against the live system on xenon where that could be done without changing anything: service status, DNS answers and web server configuration. Existing design documents were used as leads and checked against the code, not copied.

Statements marked `INFERRED` were worked out from the code but not seen in a running browser. Everything else was read directly from the code or the live system. Screens are drawn as simplified diagrams, not screenshots, so they stay correct when colours or spacing change.

E.2 Secrets

This manual contains no passwords, tokens, keys or configuration file contents. Where a credential is involved, only the file path or the service that holds it is named.

E.3 Colophon

A single self-contained HTML file with inline styles, scripts, diagrams and fonts. It makes no network requests. The PDF is rendered from the same file in headless Chromium on US Letter paper.

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